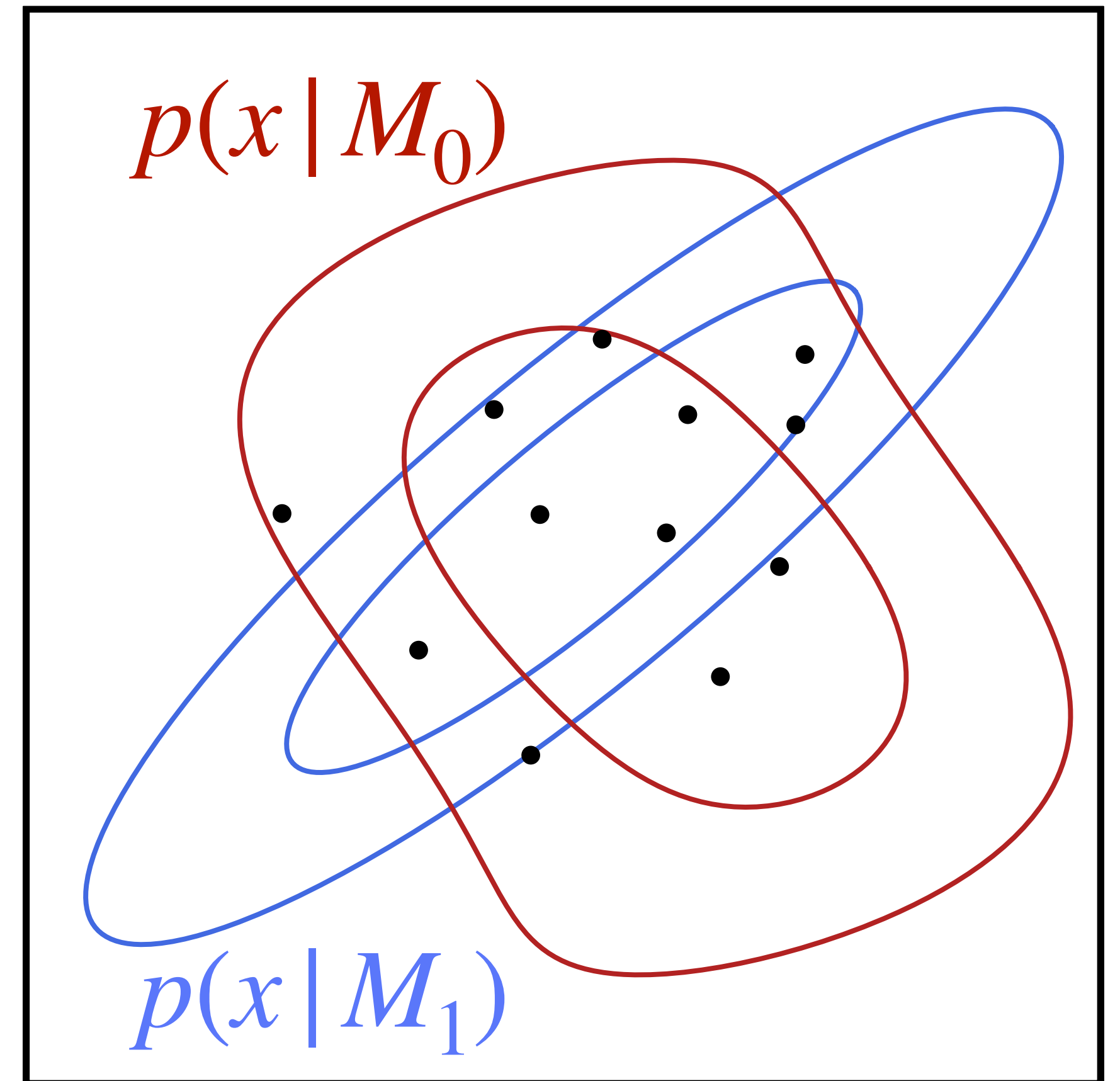


Hypothesis testing & model selection

Jed Homer LMU/ODSL



What have we done so far?

- So far we have run analyses of a mean μ and interpreted our (estimate, upper, lower).
 - i.e. we have measured $\hat{x}$, calculated $\hat{\mu}$ and calculated e.g. $\mu \in [\hat{\mu} - \sigma, \hat{\mu} + \sigma]$
 - We asked, what is μ and how accurately can we estimate it with $\hat{\mu}$?

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- How confident are we that our model $\hat{p}(\hat{\mu} \mid \mu)$ represents the data-generating process $p(\hat{\mu} \mid \mu)$?

Bayesian & Frequentist views

- How confident are we that our model $\hat{p}(\hat{\mu} | \mu)$ represents the data-generating process $p(\hat{\mu} | \mu)$?

Frequentist

- Are the measurements unusually inconsistent with a specified mean?

Hypothesis testing

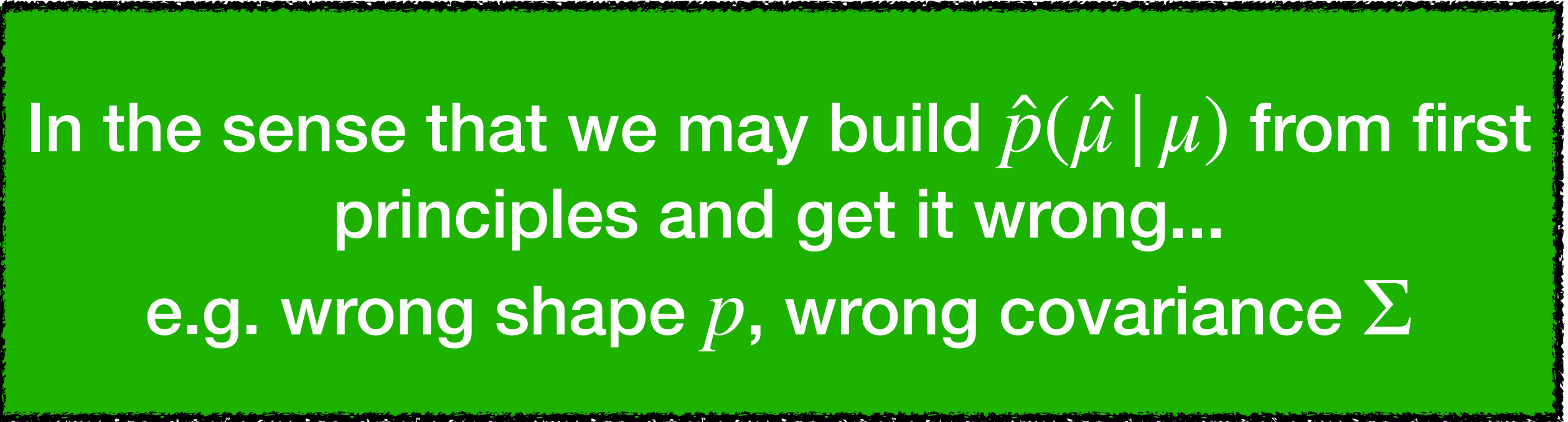
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Hypothesis testing



In the sense that we may build $\hat{p}(\hat{\mu} | \mu)$ from first principles and get it wrong...
e.g. wrong shape p , wrong covariance Σ

Bayesian & Frequentist views

- How confident are we that our model $\hat{p}(\hat{\mu} | \mu)$ represents the data-generating process $p(\hat{\mu} | \mu)$?

Frequentist

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Hypothesis testing

Bayesian

- How should the measurements change our relative support for competing models?

Model selection

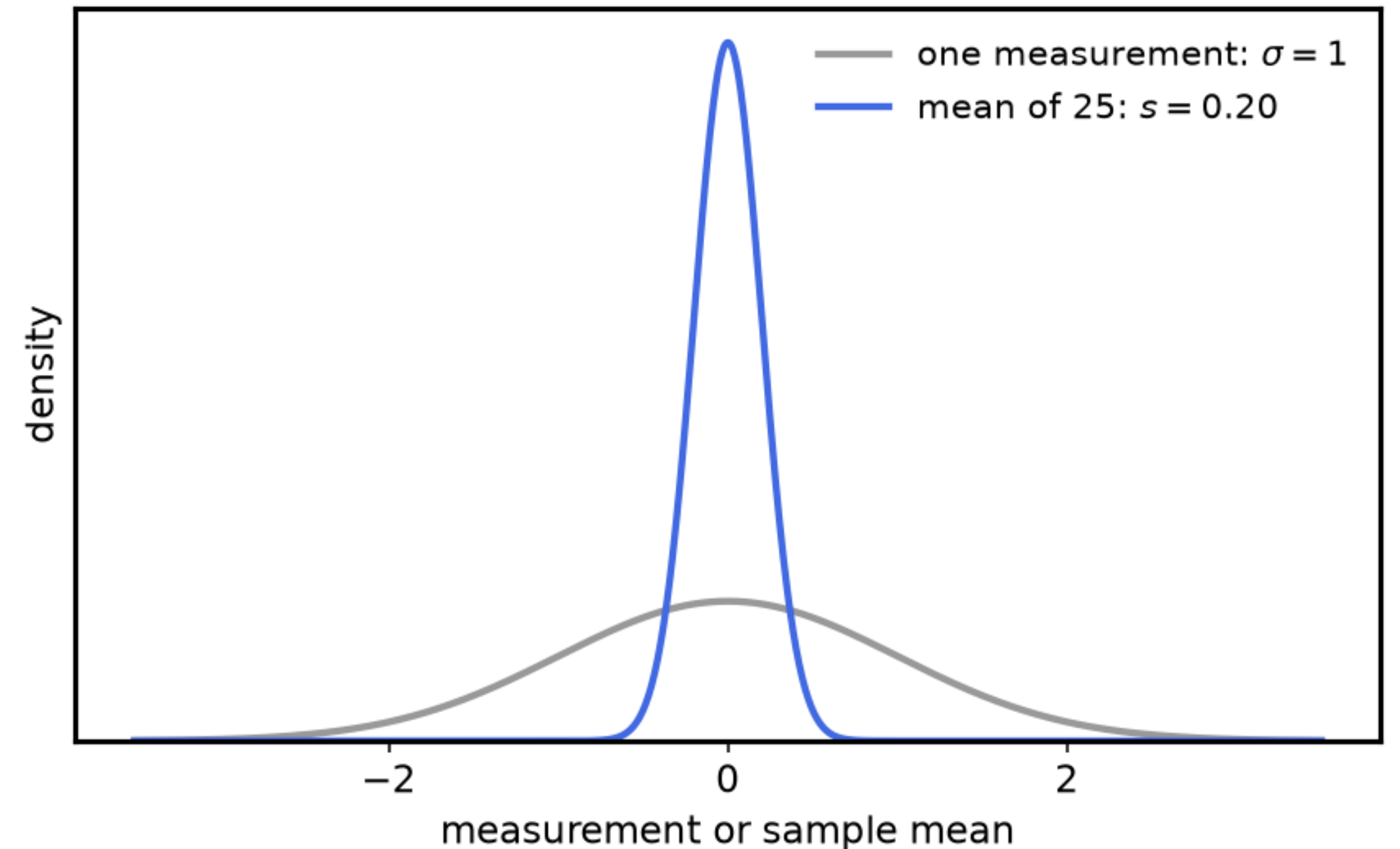
Frequentist hypothesis testing

- Let's return to our example...

$$x_i \sim \mathcal{G}[x | \mu, \Sigma]$$

$$\hat{\mu} = \bar{x}$$

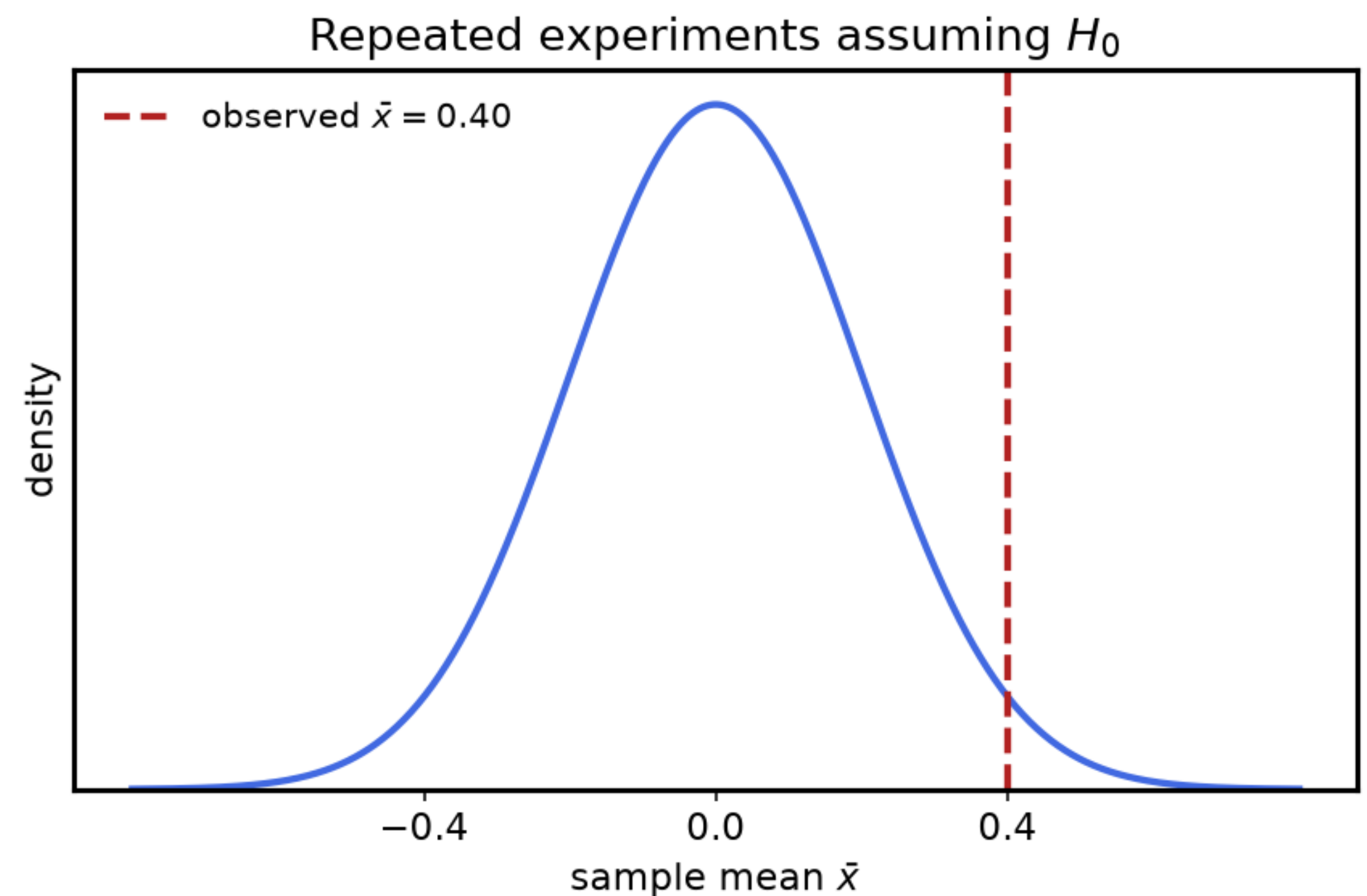
$$\bar{x} \sim \mathcal{G}[\bar{x} | \mu, n^{-1}\Sigma]$$



Hypothesis about our model $p(\hat{\mu} \mid \mu)$

- Hypotheses regarding μ

$$H_0 : \mu = \mu_0, \quad H_1 : \mu \neq \mu_0.$$

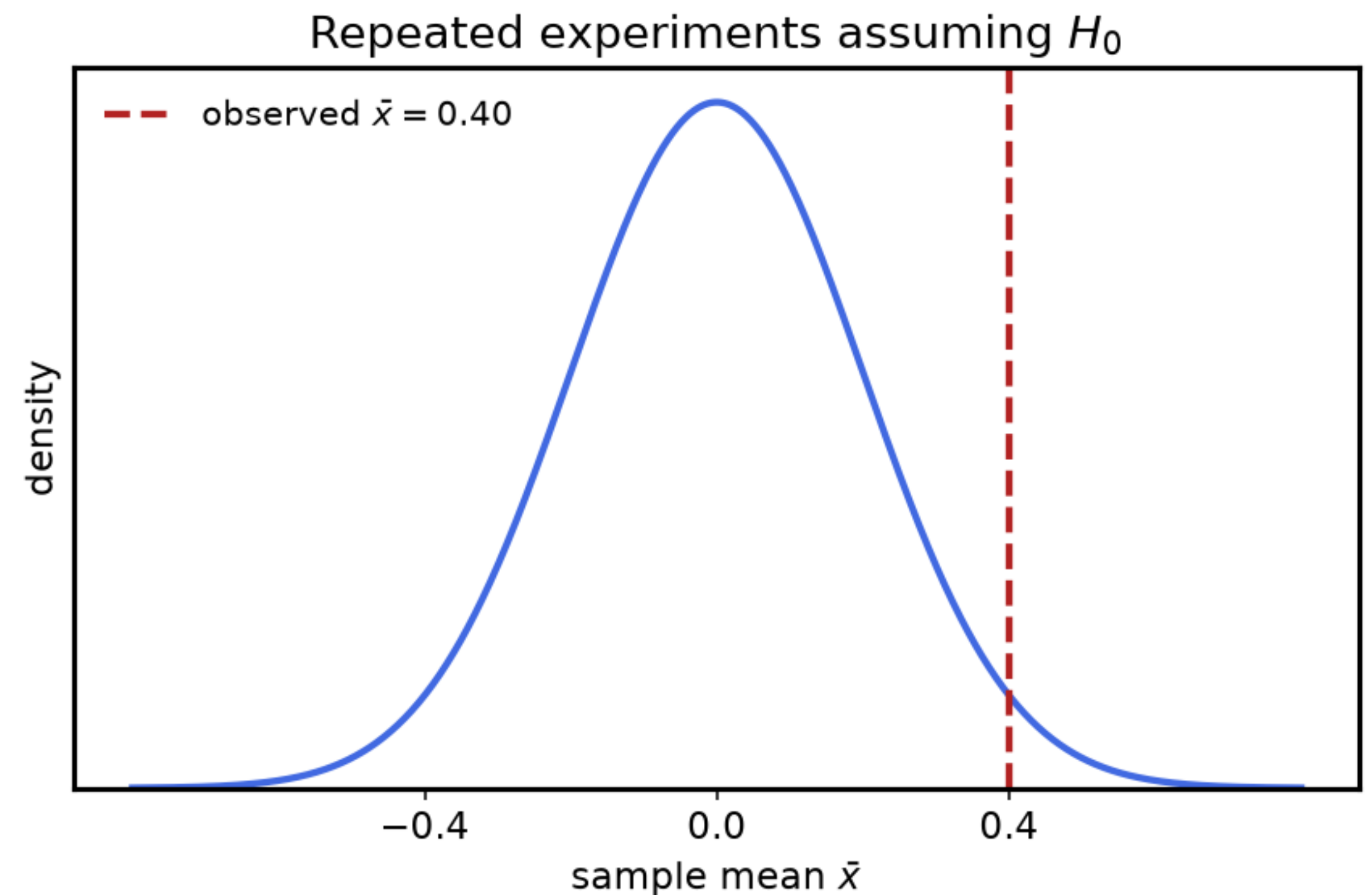


Hypothesis about our model $p(\hat{\mu} | \mu)$

- Hypotheses regarding μ

$$H_0 : \mu = \mu_0, \quad H_1 : \mu \neq \mu_0.$$

- H_1 allows for shift of μ around μ_0 ,
- H_0 asserts a fixed value μ_0 ,
- Both hypotheses assume noise σ^2 .

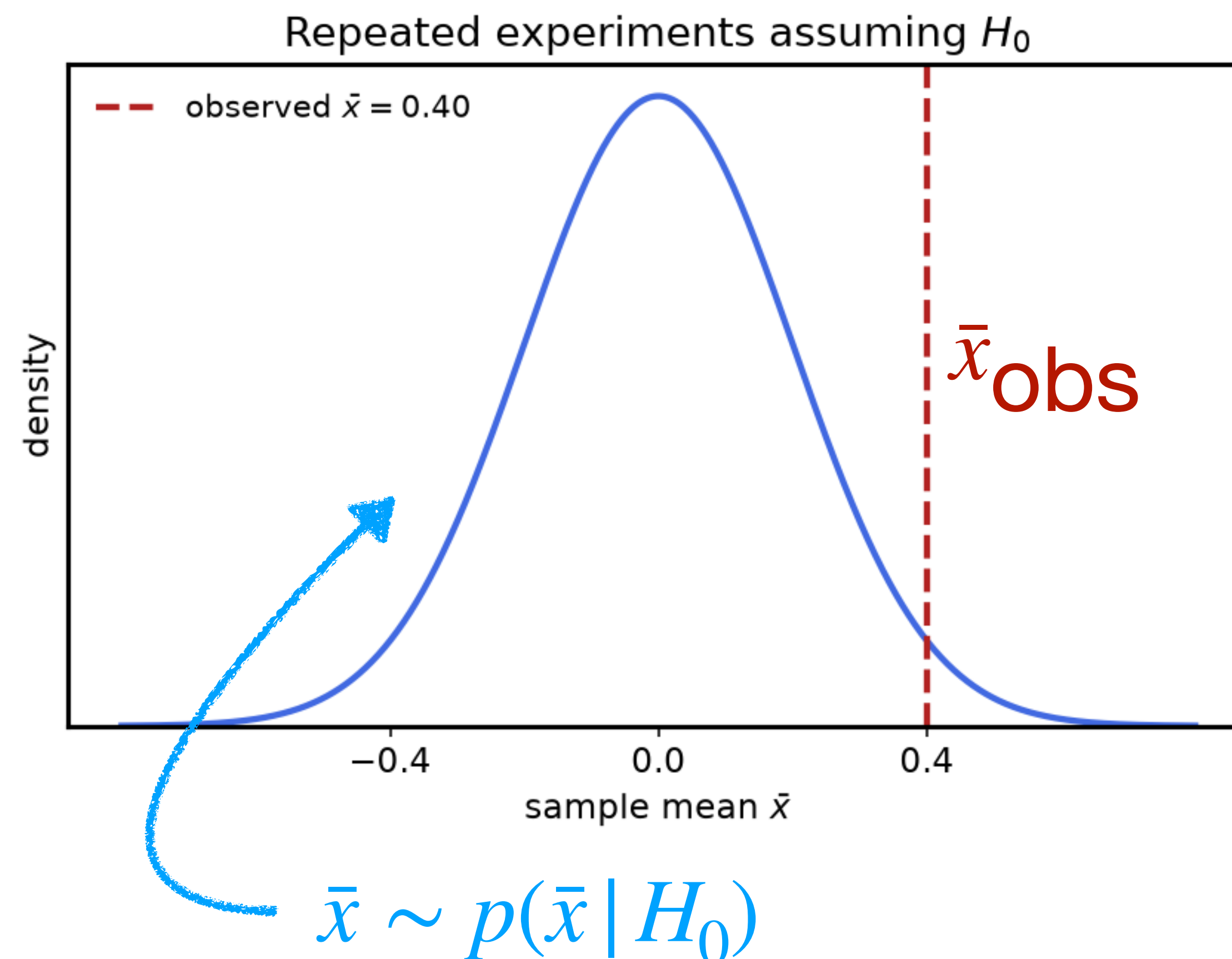


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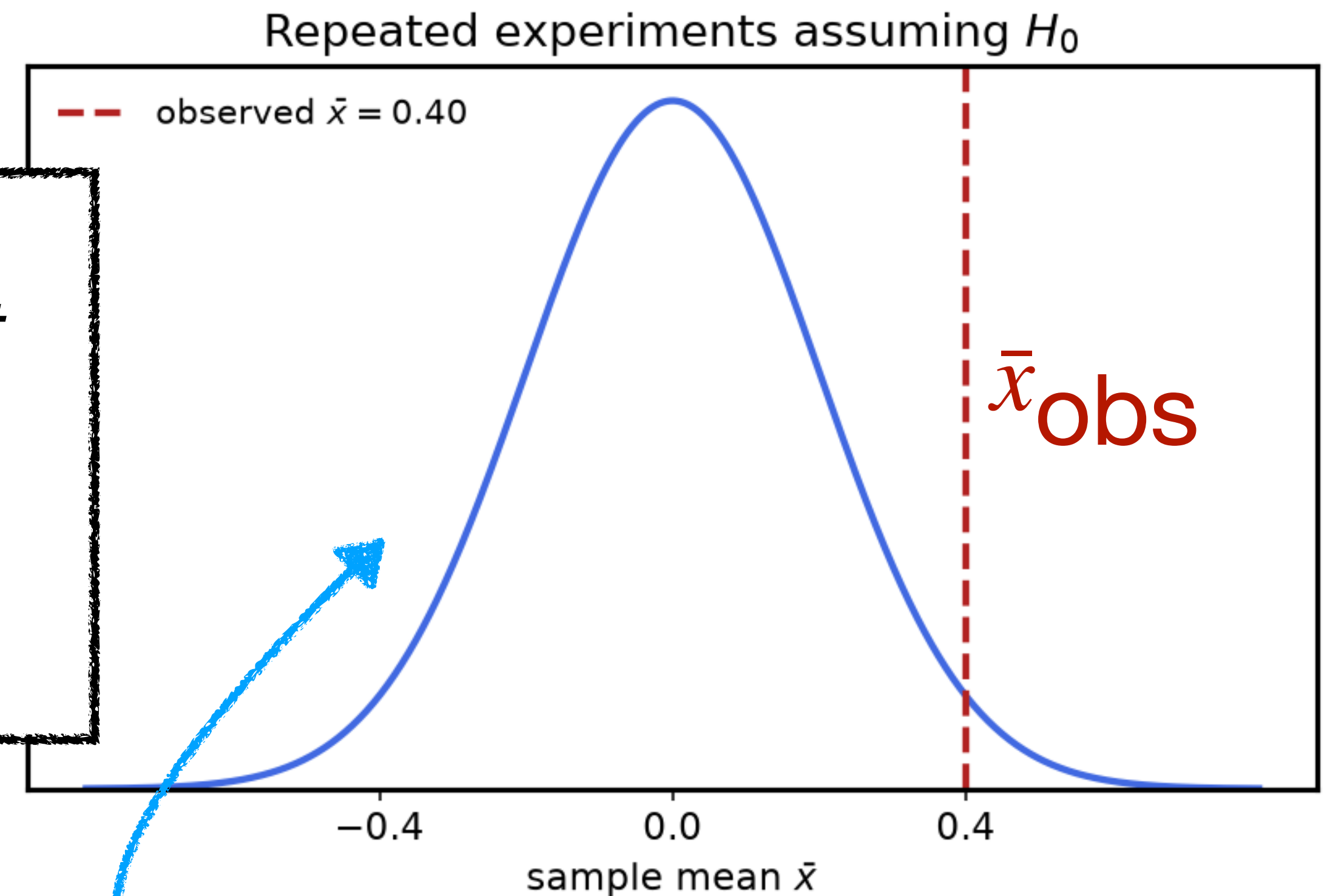
- Let's say we measure $\bar{x}_{\text{obs}}$...



Hypothesis about our model $p(\hat{\mu} | \mu)$

- Hypotheses regarding μ

- What's the probability of obtaining $\bar{x}$ *at least as far from μ_0 as $\bar{x}_{\text{obs}}$* , assuming $\mu = \mu_0$?



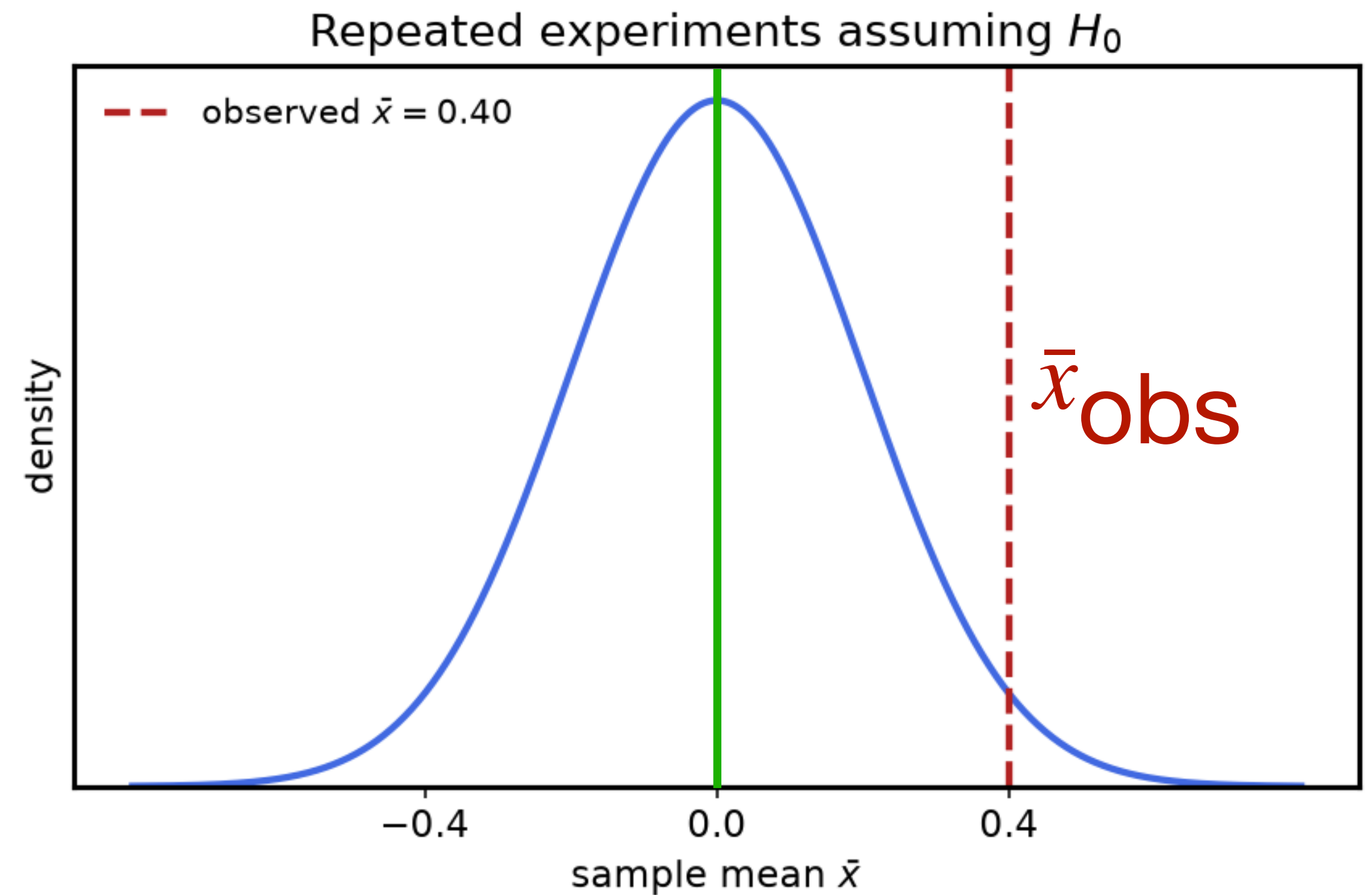
$$\bar{x} \sim p(\bar{x} | H_0)$$

p -values

- Given Gaussian data, we can standardise this question.
- Define

$$z = \frac{\bar{x} - \mu_0}{\sigma^2}$$

$$z \sim \mathcal{G}[z | 0, 1]$$

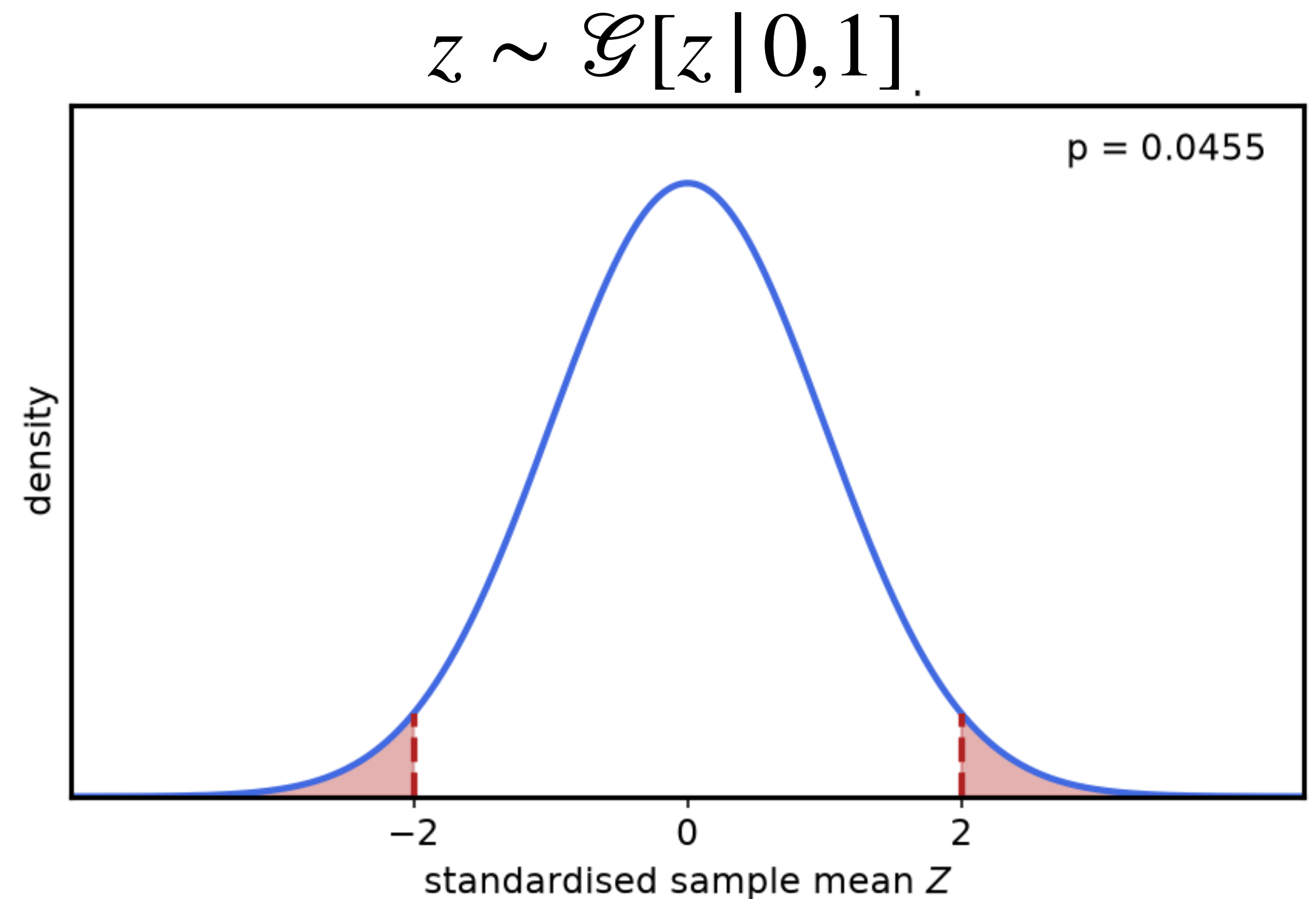


p -values

- How likely is it to scatter more than z_{obs} ?

$$p = \Pr[|z| \geq |z_{\text{obs}}| \mid H_0]$$

- This is a p -value, be careful!

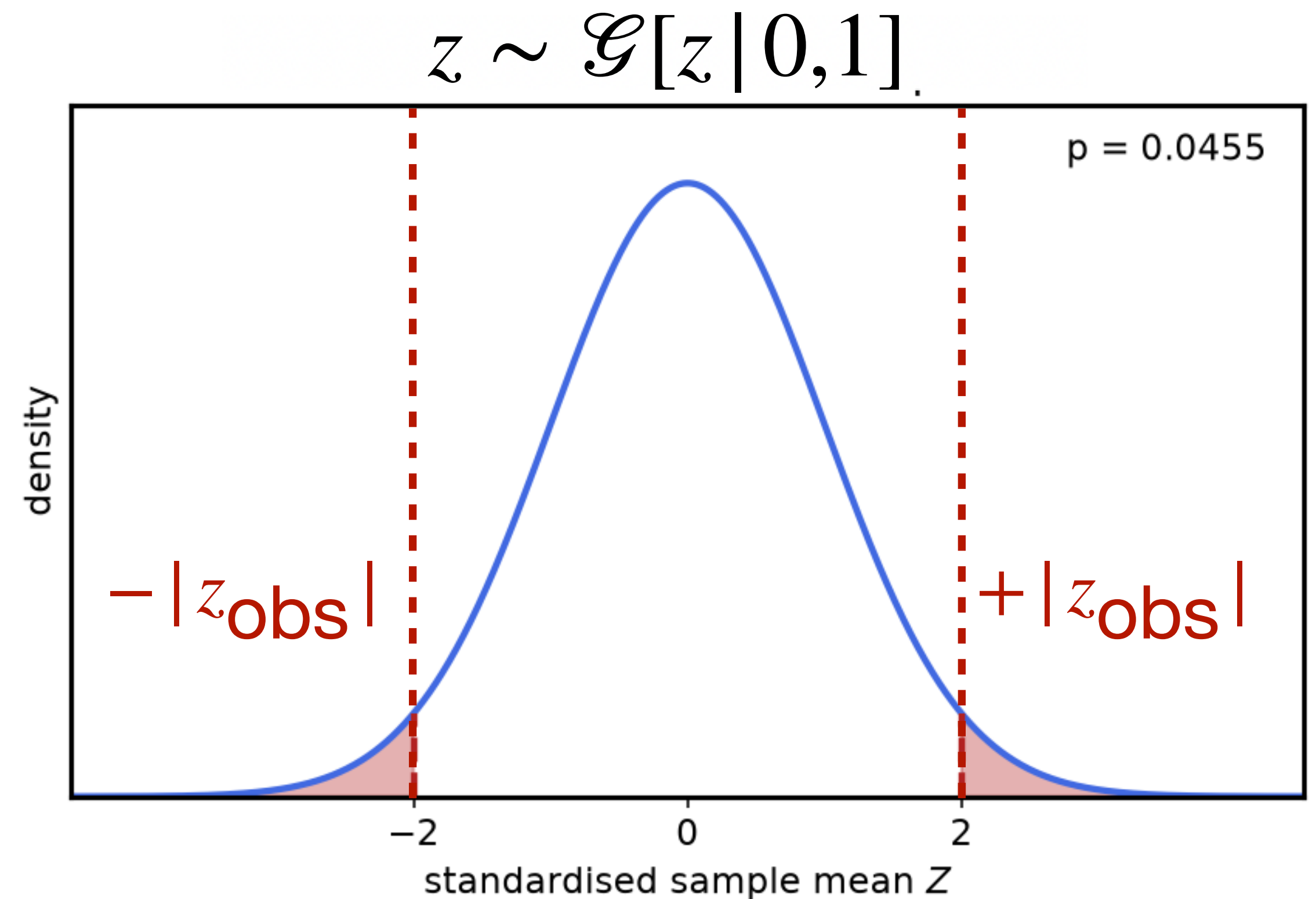


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p -values

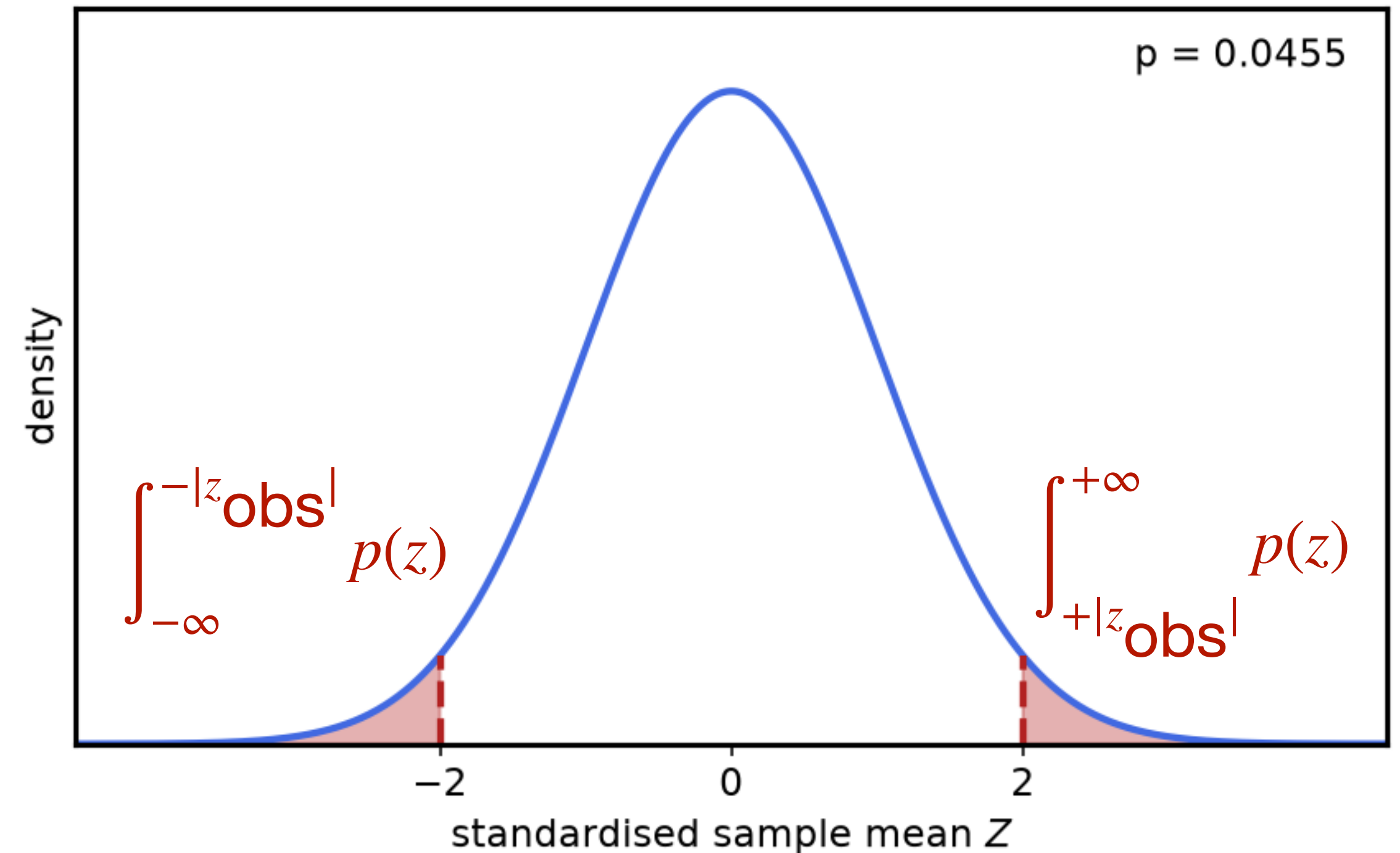
- How likely is it to scatter more than z_{obs} ?

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- Calculate p -value as

$$p = \int_{-\infty}^{-|z_{\text{obs}}|} p(z) + \int_{+|z_{\text{obs}}|}^{+\infty} p(z)$$

$$z \sim \mathcal{G}[z \mid 0, 1]$$



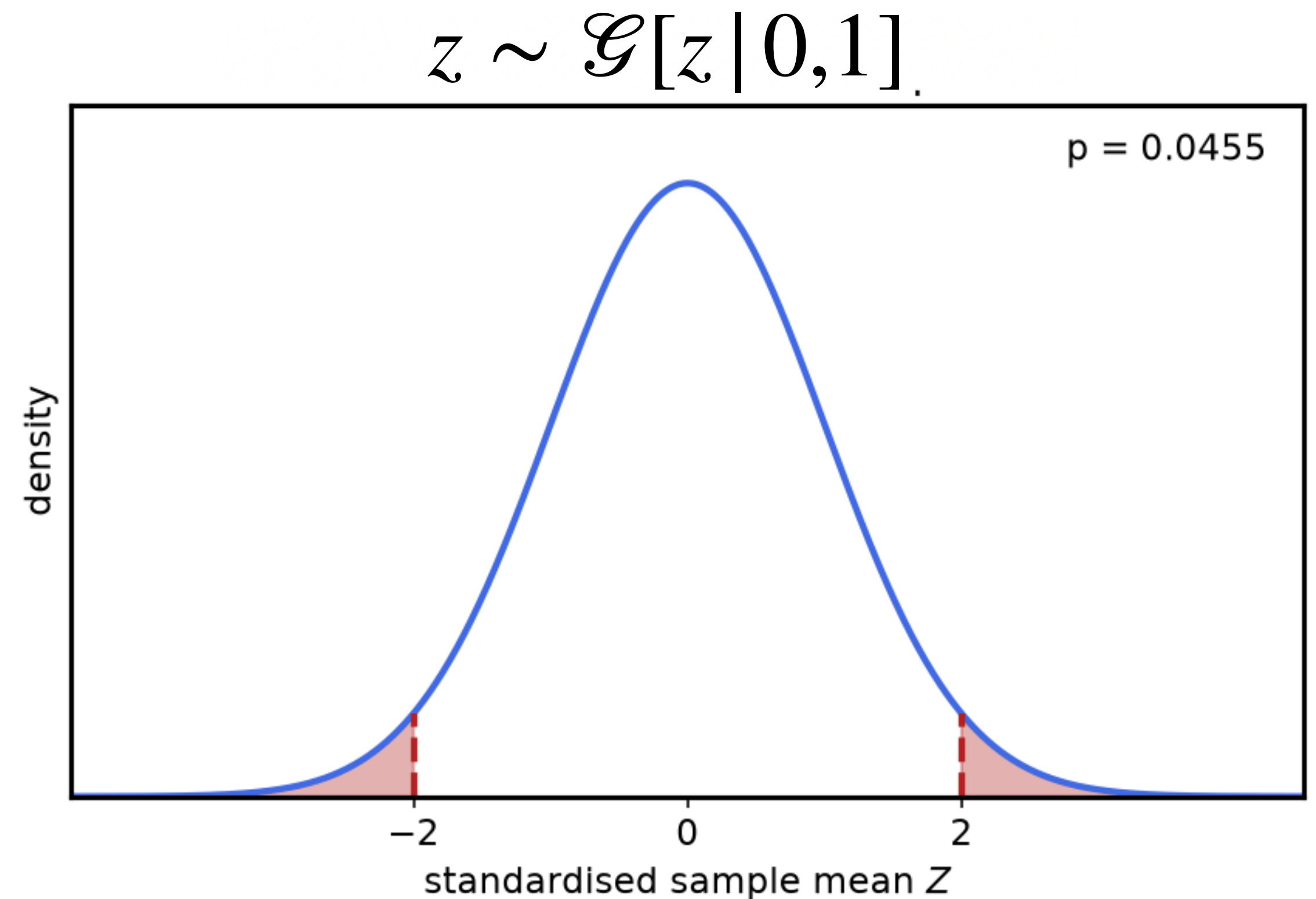
p -values

- How likely is it to scatter more than z_{obs} ?

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$$p = 2[1 - \text{CDF}(|z_{\text{obs}}|)]$$



p -values

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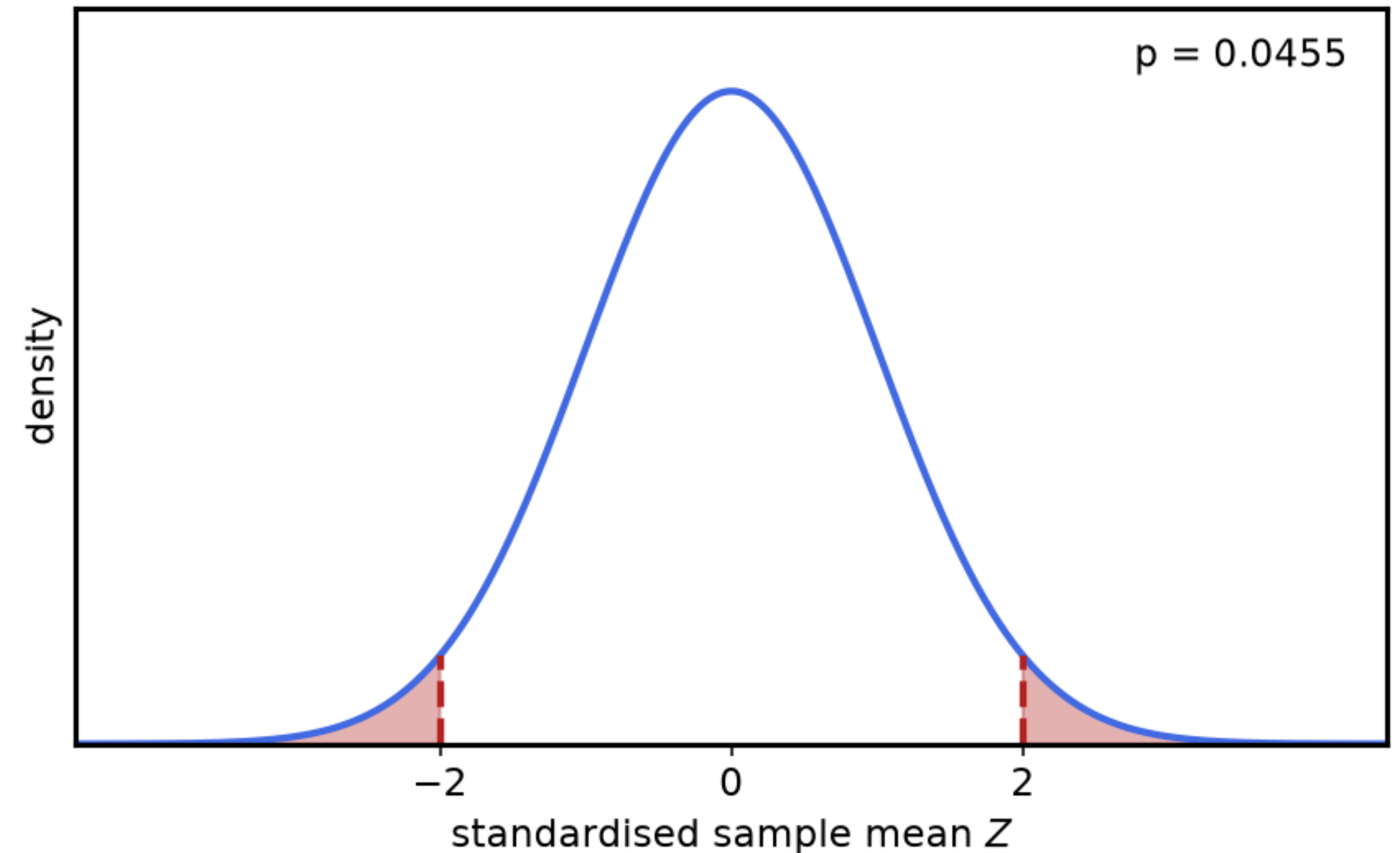
- Calculate p -value as

$$p = 2[1 - \text{CDF}(|z_{\text{obs}}|)]$$



This has used the symmetry of our Gaussian model (with H_0 or H_1) around the mean!

$$z \sim \mathcal{G}[z \mid 0, 1]$$



p -values

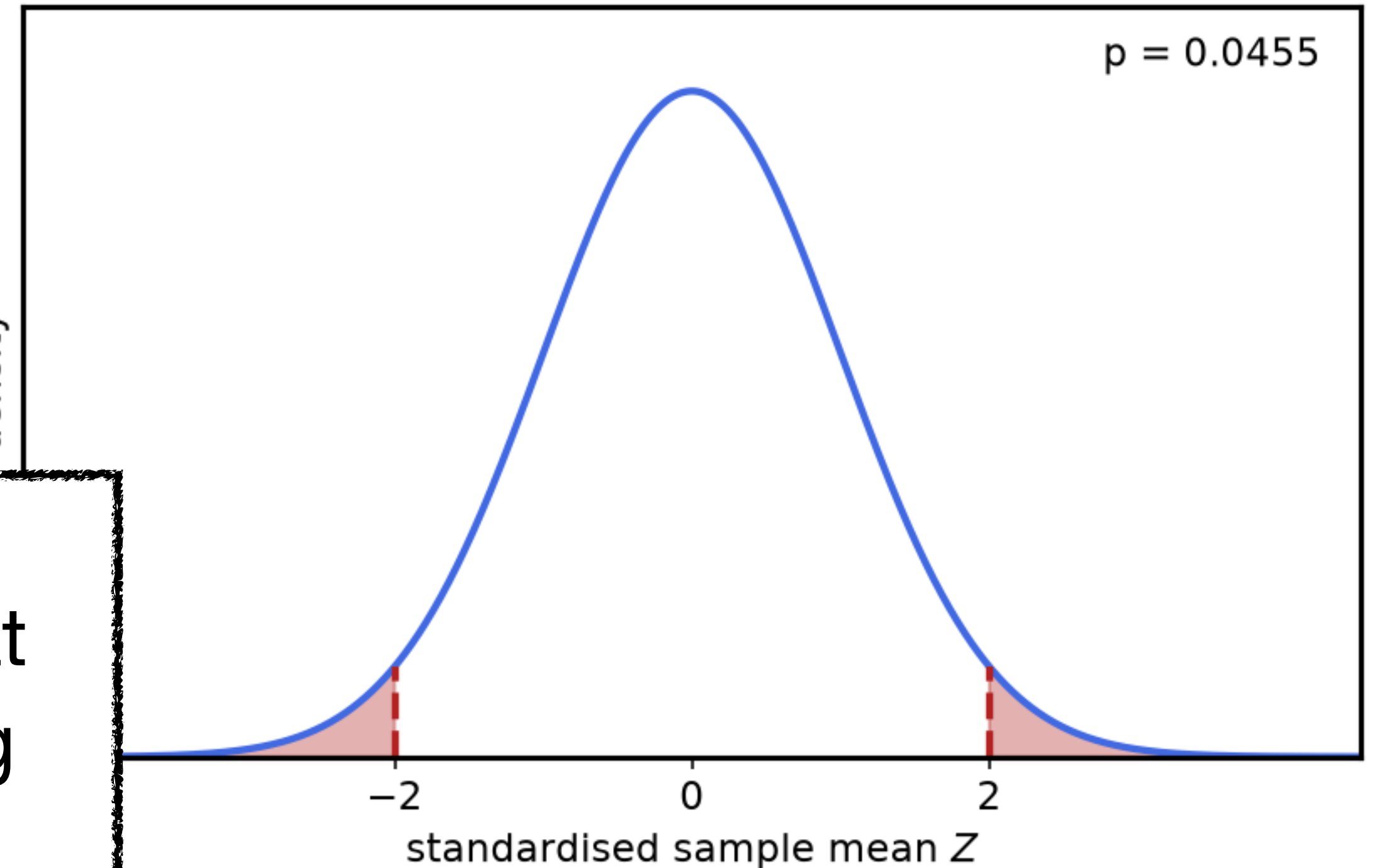
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density

- What's the probability of obtaining $\bar{x}$ at least as far from μ_0 as $\bar{x}_{\text{obs}}$, assuming $\mu = \mu_0$?



p -values

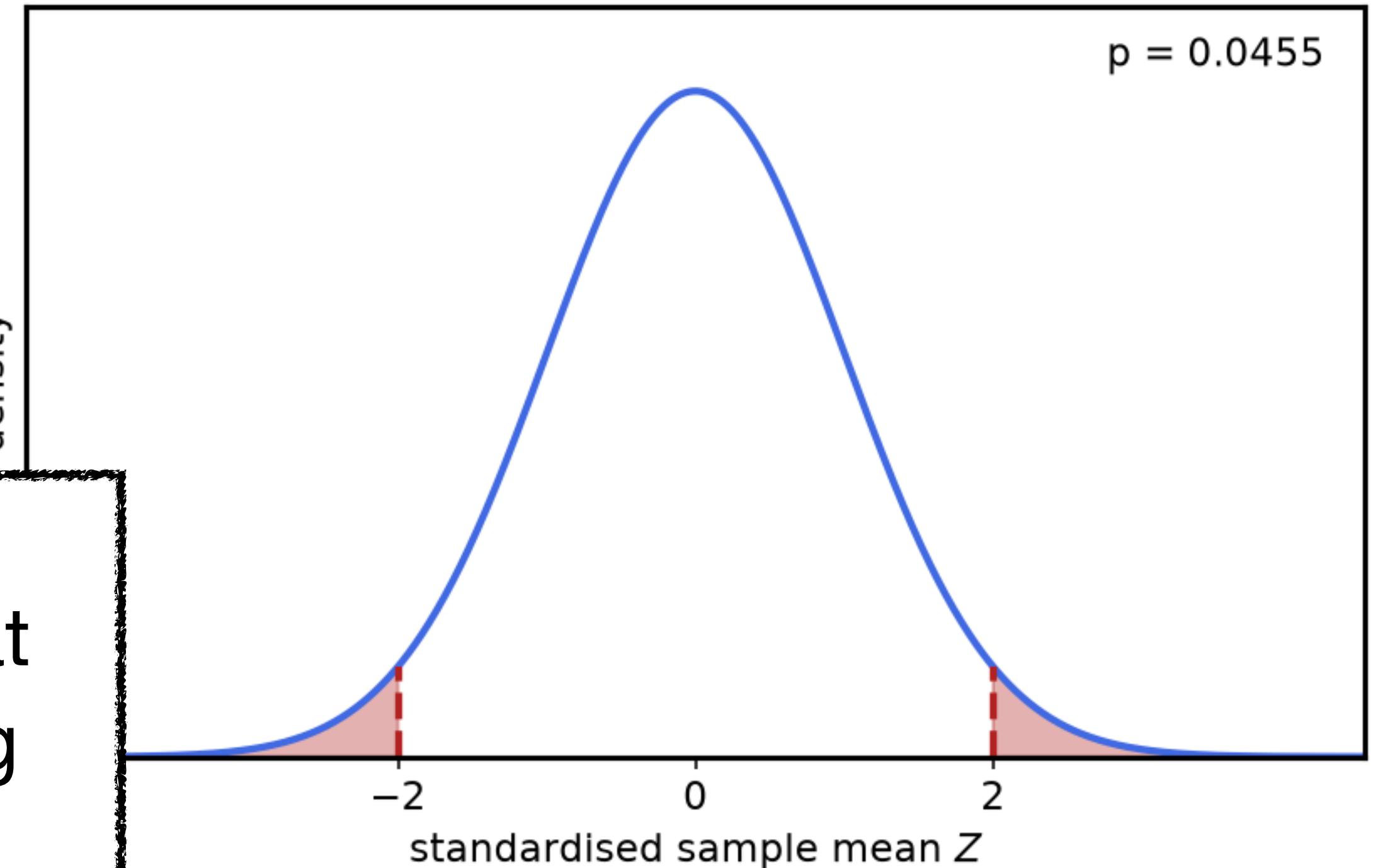
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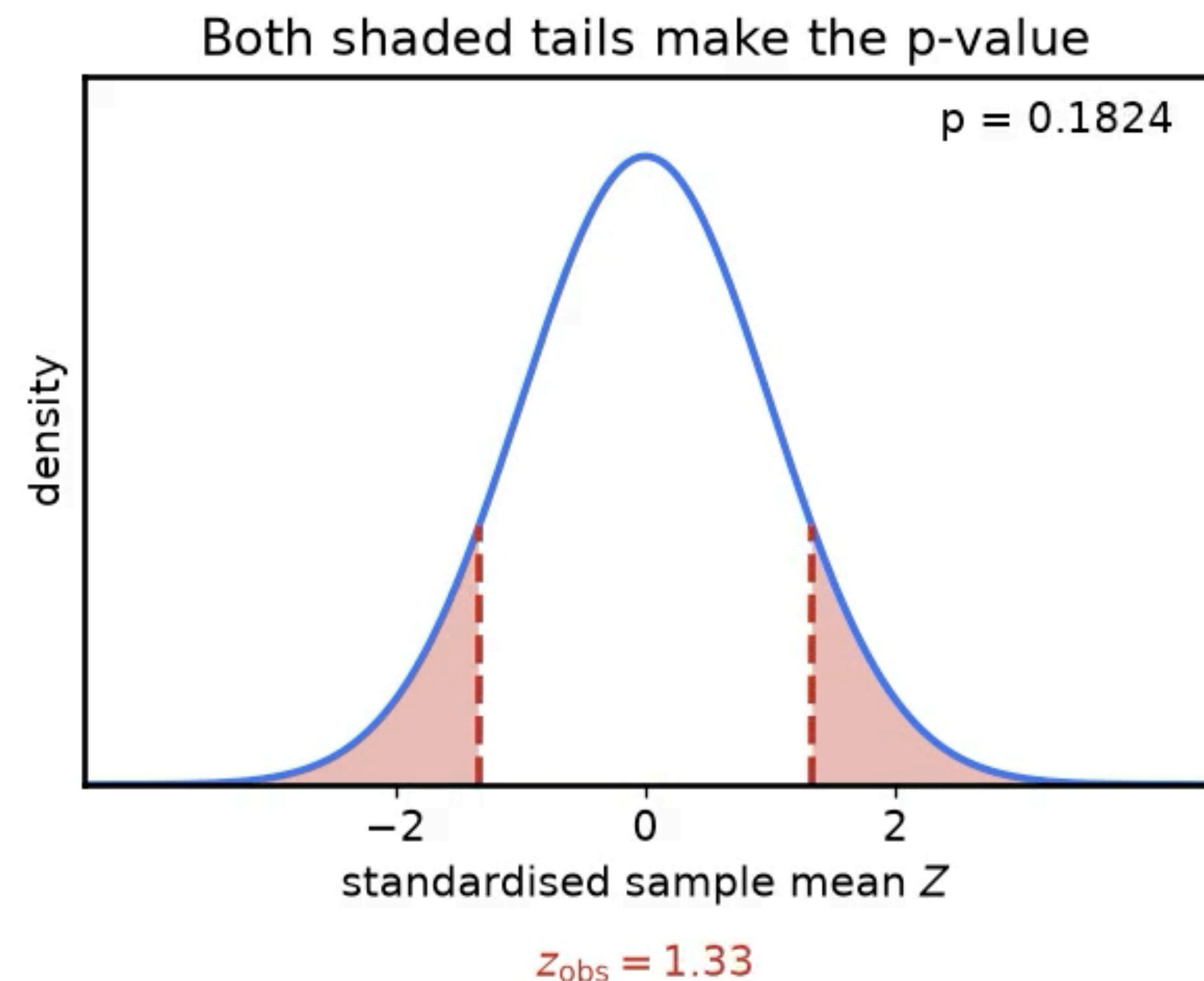
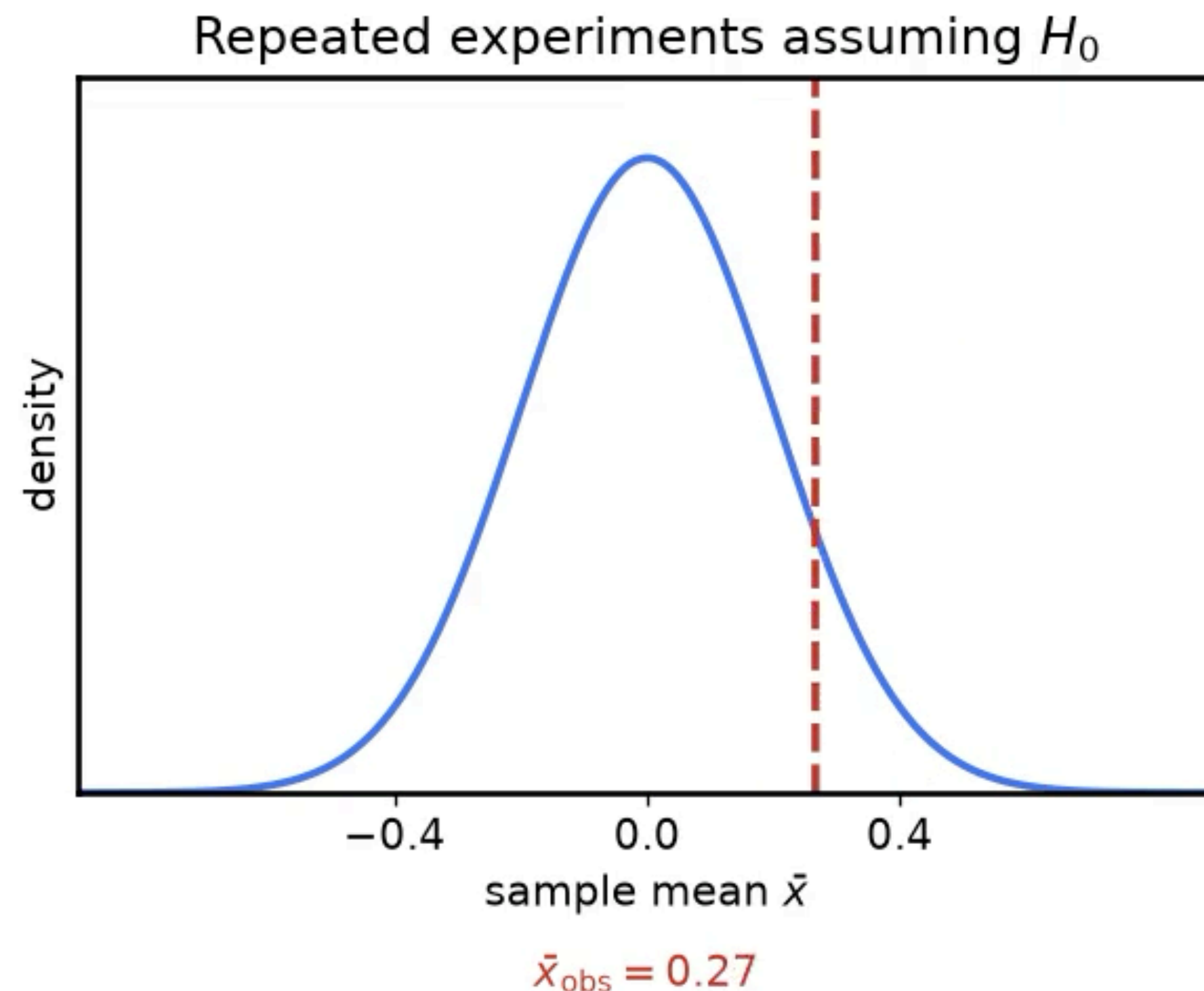
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- What's the probability of obtaining $\bar{x}$ at least as far from μ_0 as $\bar{x}_{\text{obs}}$, assuming $\mu = \mu_0$?



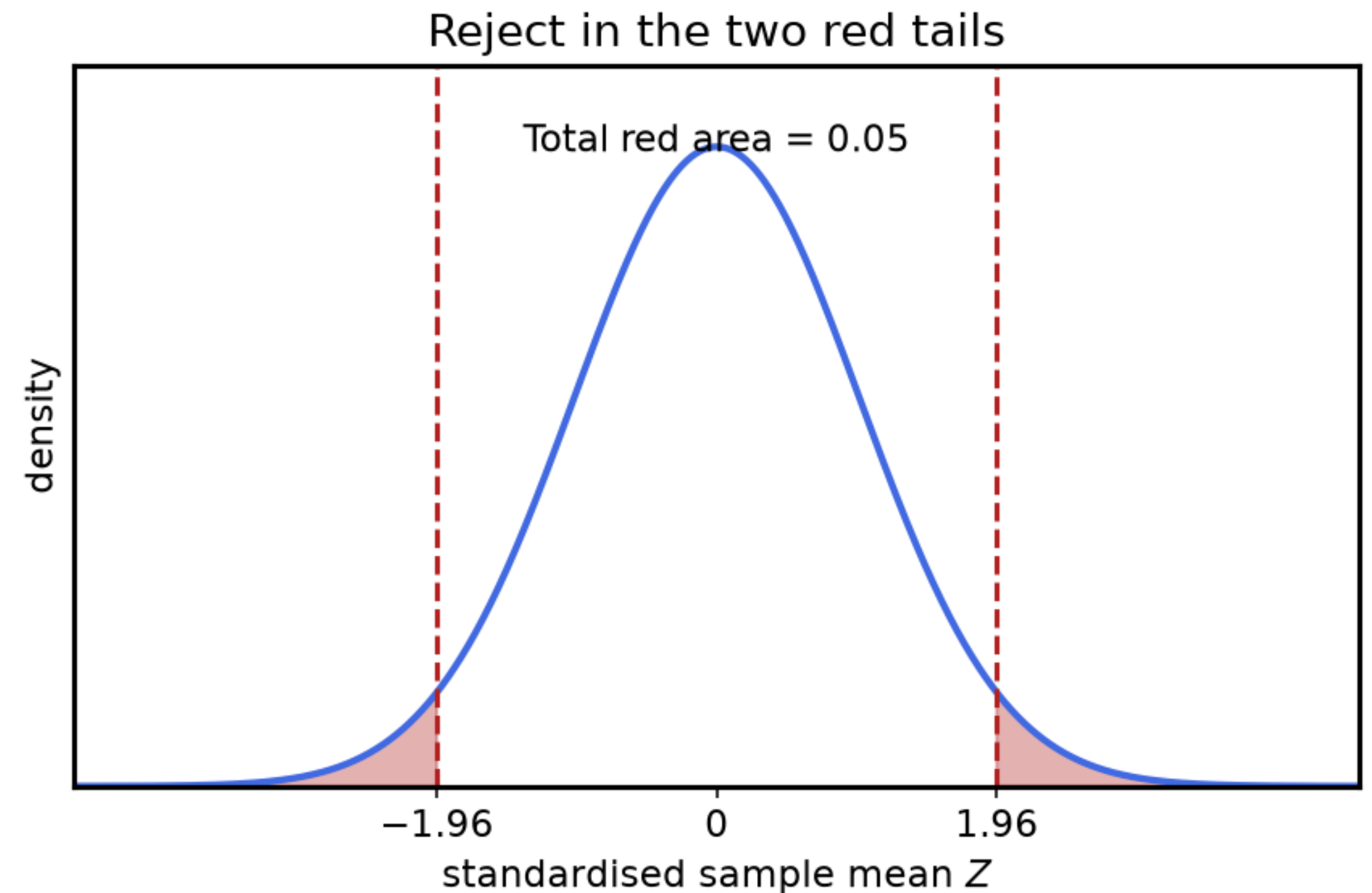
Example: p -values for repeated experiments

$$p = \Pr[|z| \geq |z_{\text{obs}}| \mid H_0]$$



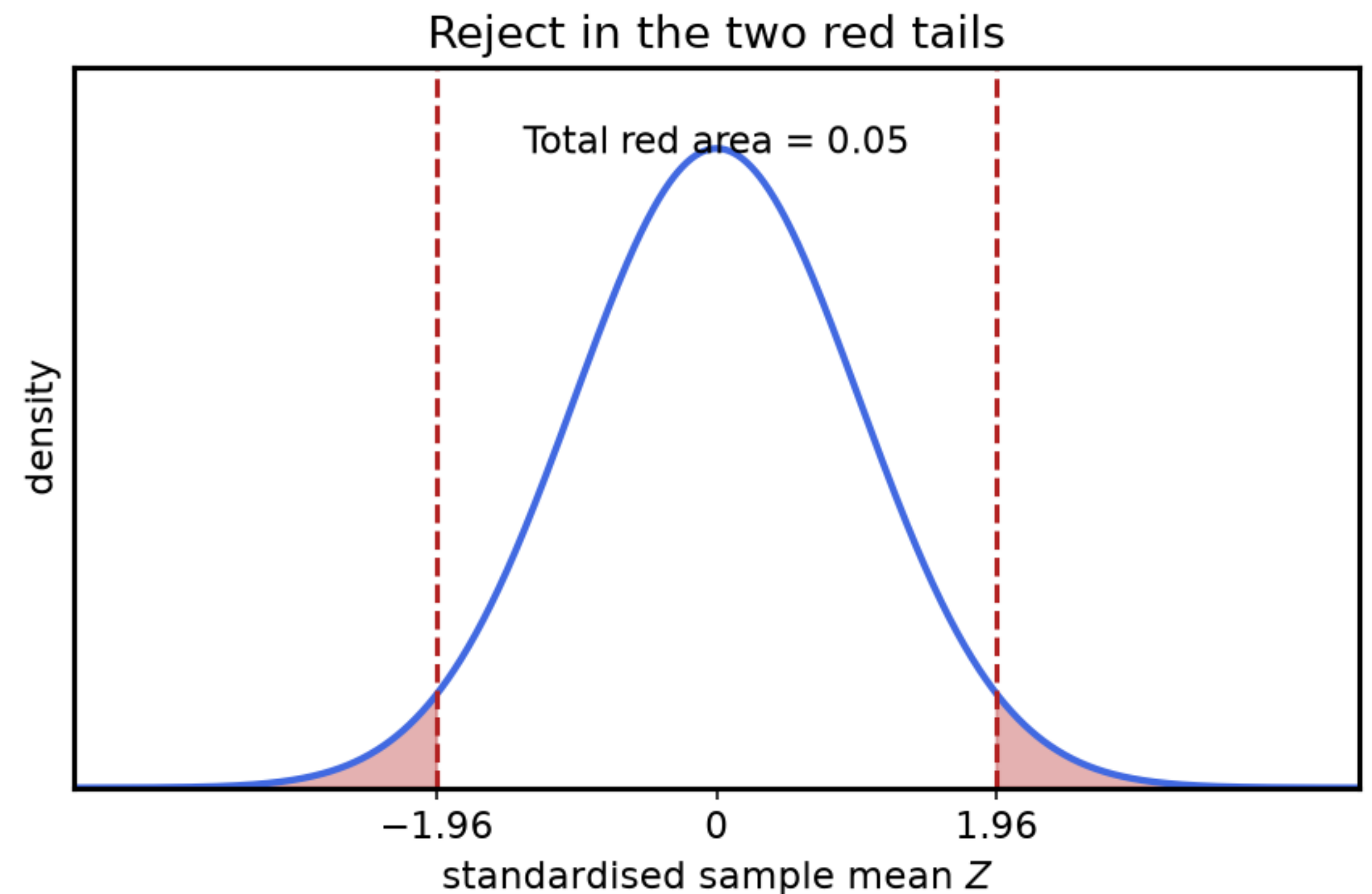
From a p -value to a decision

- Why would we **reject** the hypothesis H_0 ?
- In many repeated experiments, if it is not likely that the scatter of $\hat{\mu}$ around μ_0 is observed, we should reject H_0



From a p -value to a decision

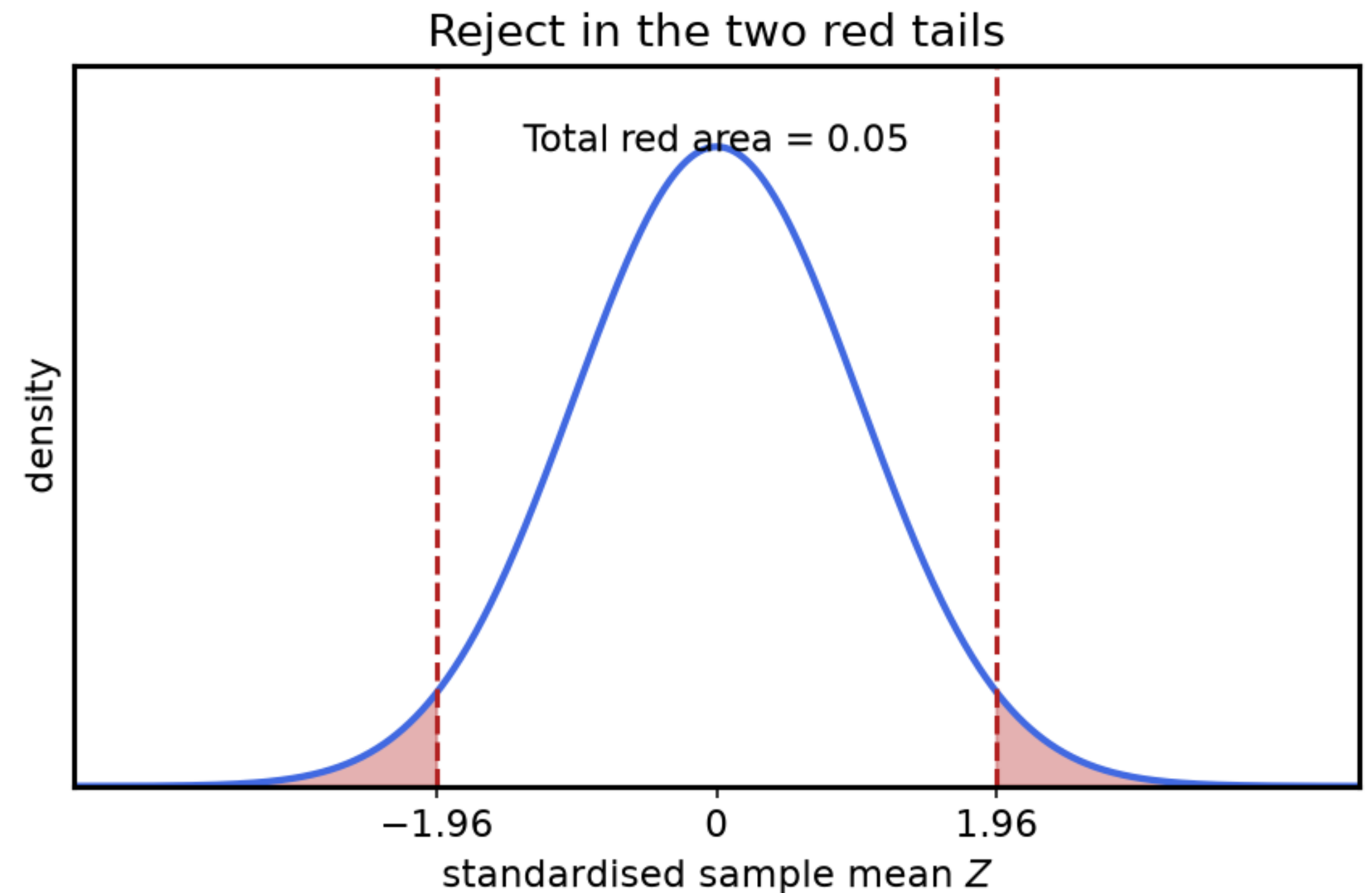
- Why would we **reject** the hypothesis H_0 ?
 - In many repeated experiments, if it is not likely that the scatter of $\hat{\mu}$ around μ_0 is observed, we should reject H_0
 - Ultimately we choose the fraction of experiments.
 - Common to choose $\alpha = 0.05$.
- $\implies$ Reject H_0 when $p < \alpha$.



From a p -value to a decision

- Choose a region $|z| > c$, where the probability given H_0 is α

$$\Pr[|z| > c \mid H_0] = \alpha$$

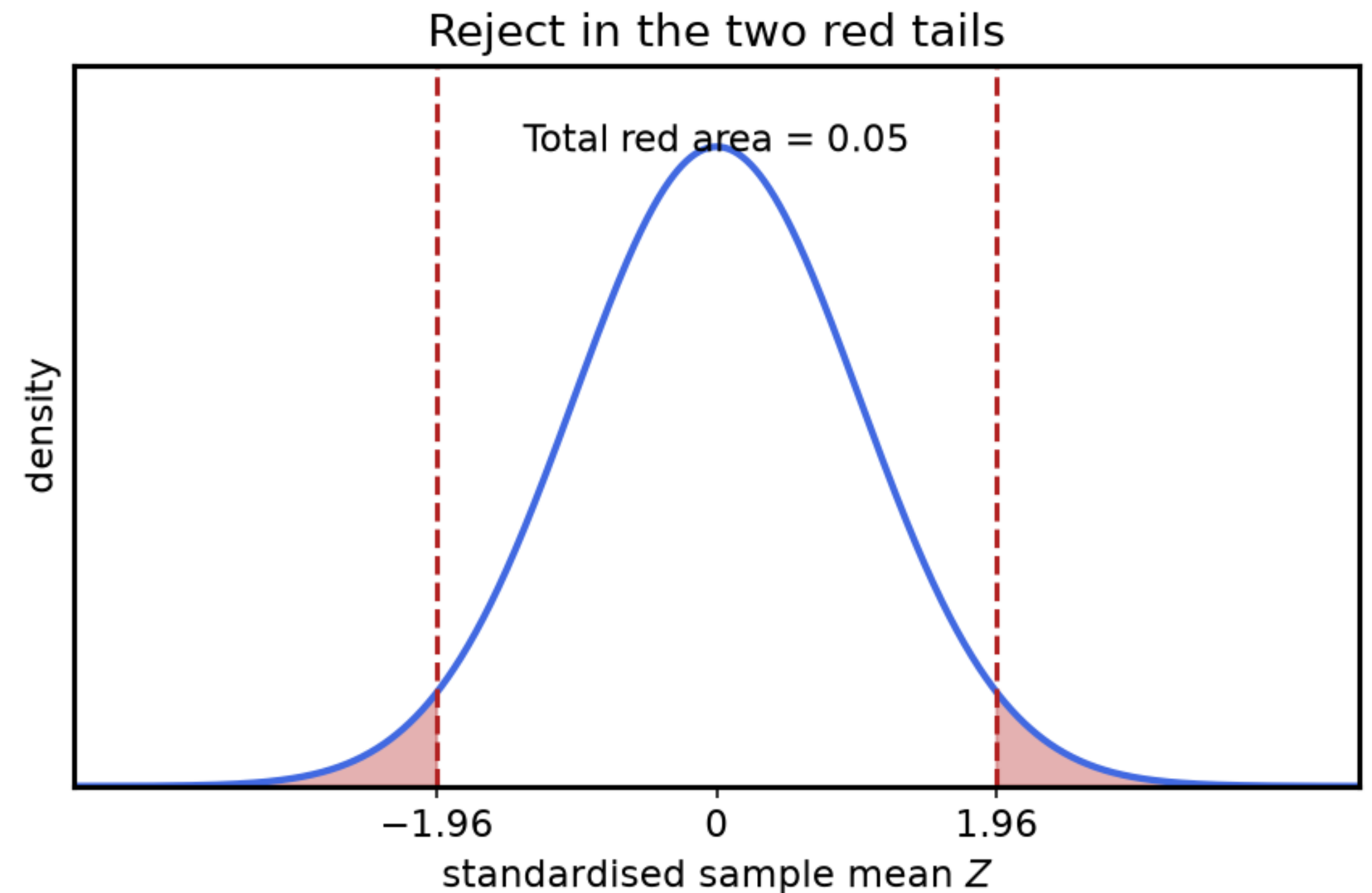


From a p -value to a decision

- Choose a region $|z| > c$, where the probability given H_0 is α

$$\Pr[|z| > c \mid H_0] = \alpha$$

c is just a threshold we set in our analysis by α !



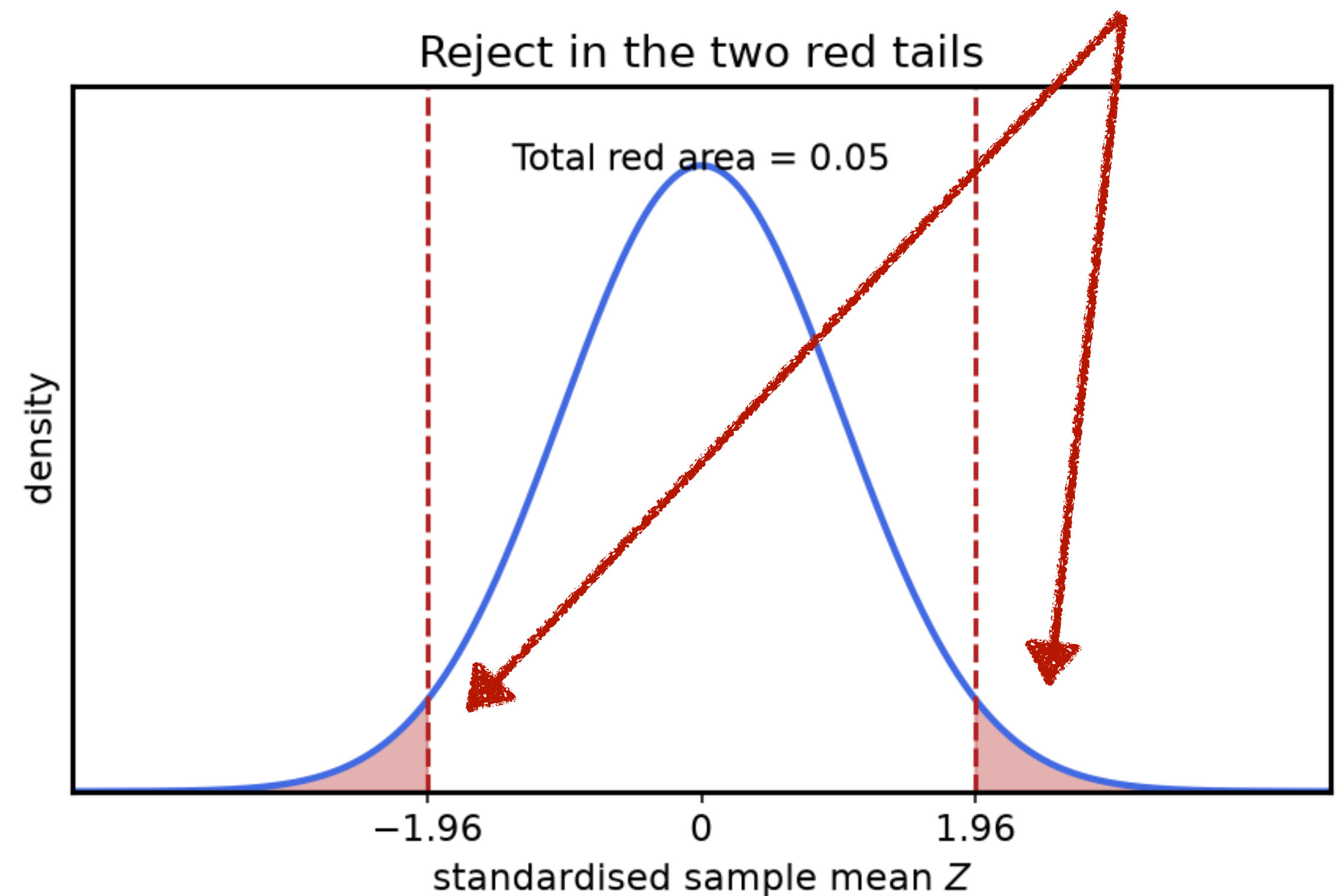
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Φ = CDF of Gaussian

$$\Phi(c) = 1 - \frac{\alpha}{2}$$



From a p -value to a decision

Φ = CDF of Gaussian

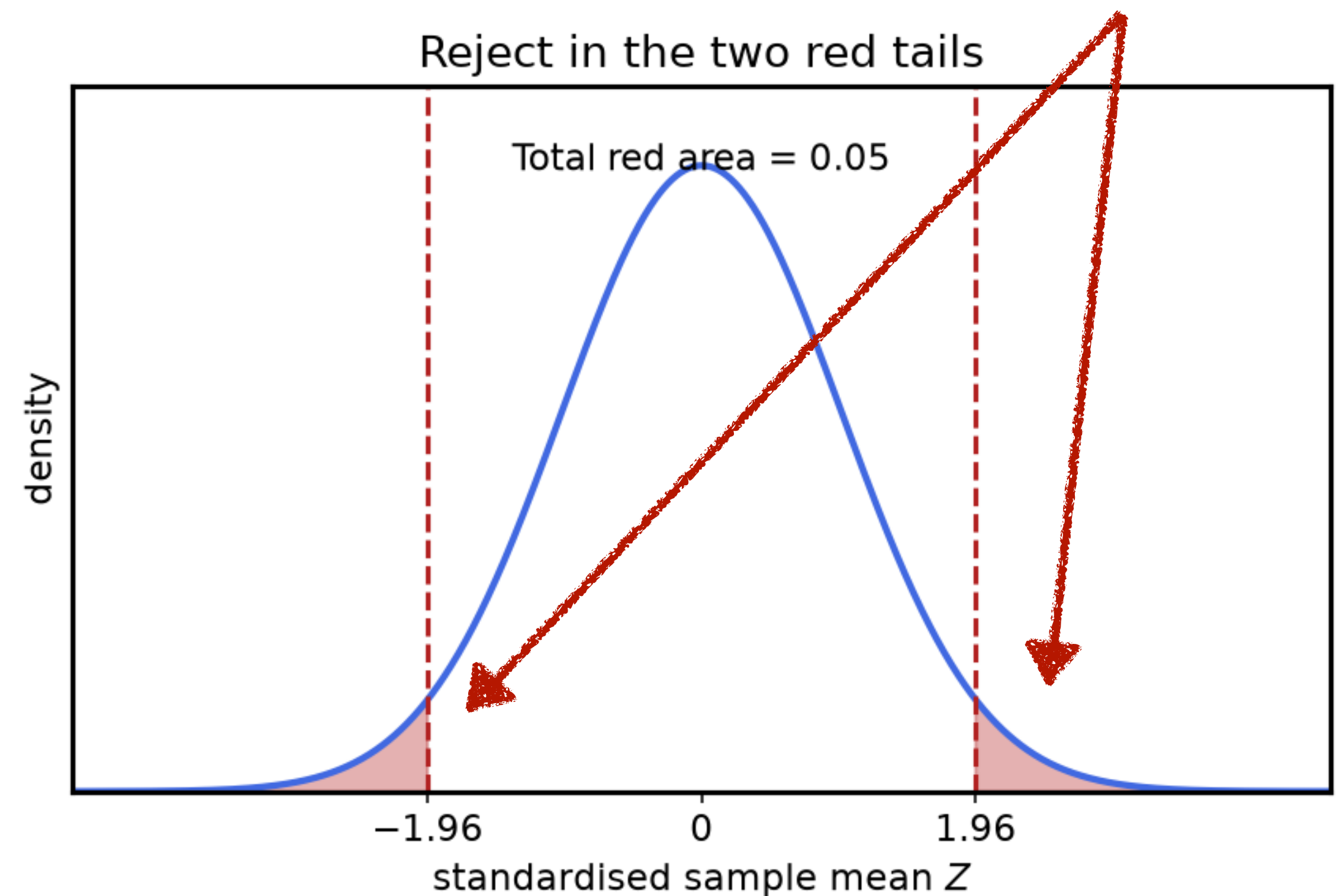
$$\Phi(c) = 1 - \frac{\alpha}{2}$$

- Choose a region $|z| > c$, where the probability given H_0 is α

$$\Pr[|z| > c \mid H_0] = \alpha$$

- Under Gaussian $p(z)$, the red area is given by the CDF

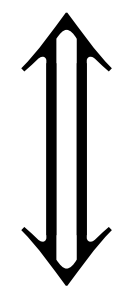
$$c = \Phi^{-1}\left(1 - \frac{\alpha}{2}\right) \approx 1.96$$



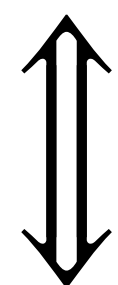
From a p -value to a decision

- We would then **reject** H_0 if

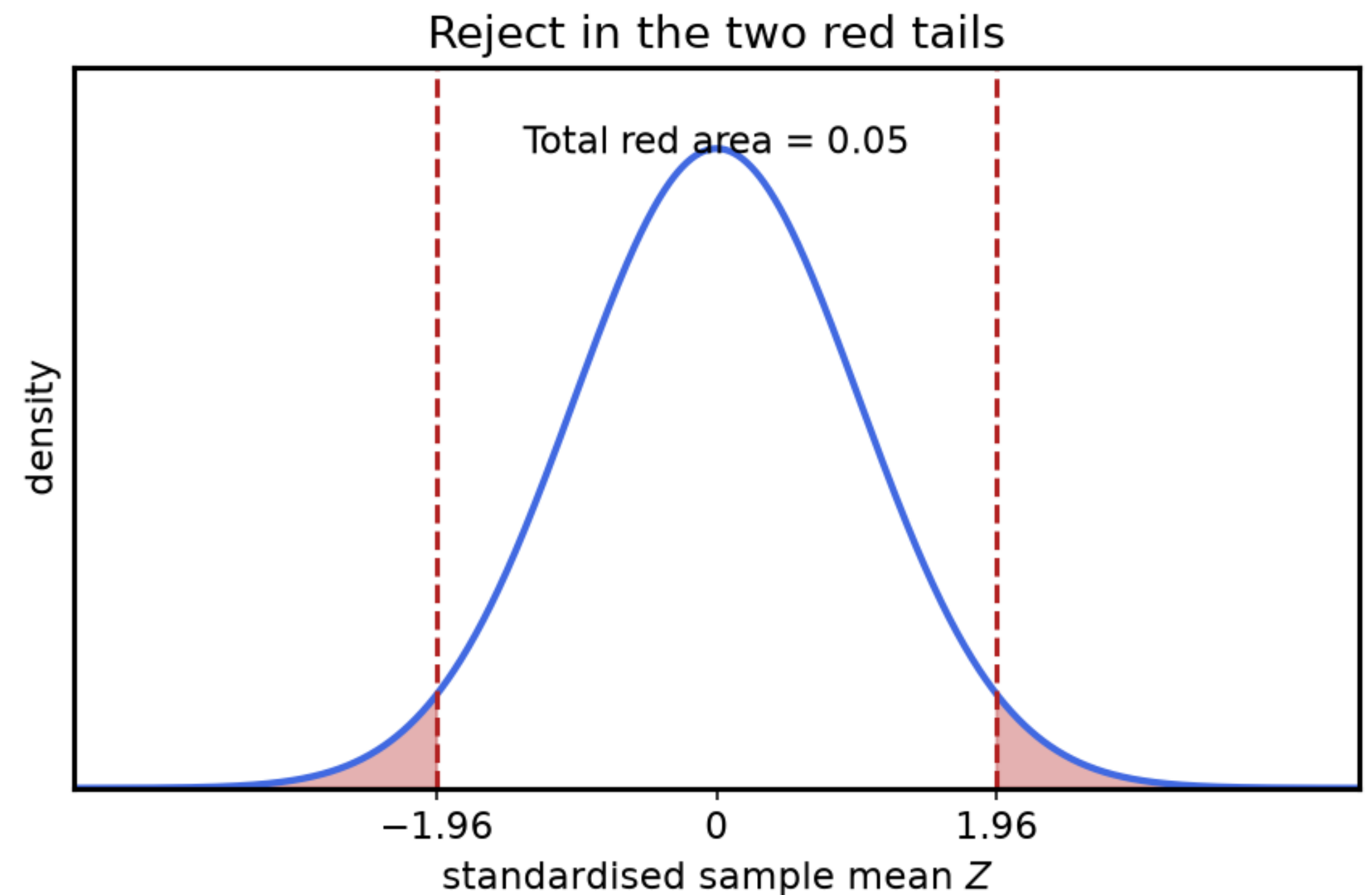
$$p < 0.05$$



$$|z_{\text{obs}}| > 1.96$$

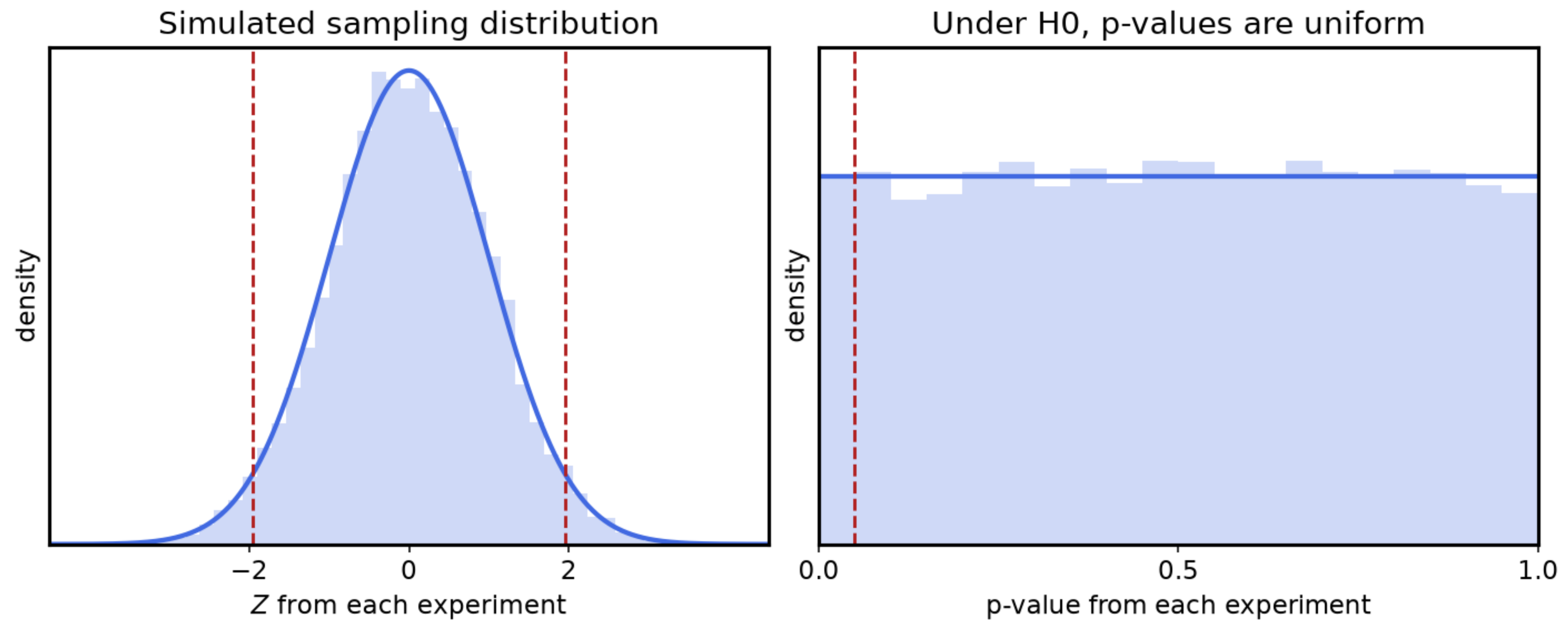


$$|\bar{x}_{\text{obs}} - \mu_0| > 1.96\sigma$$



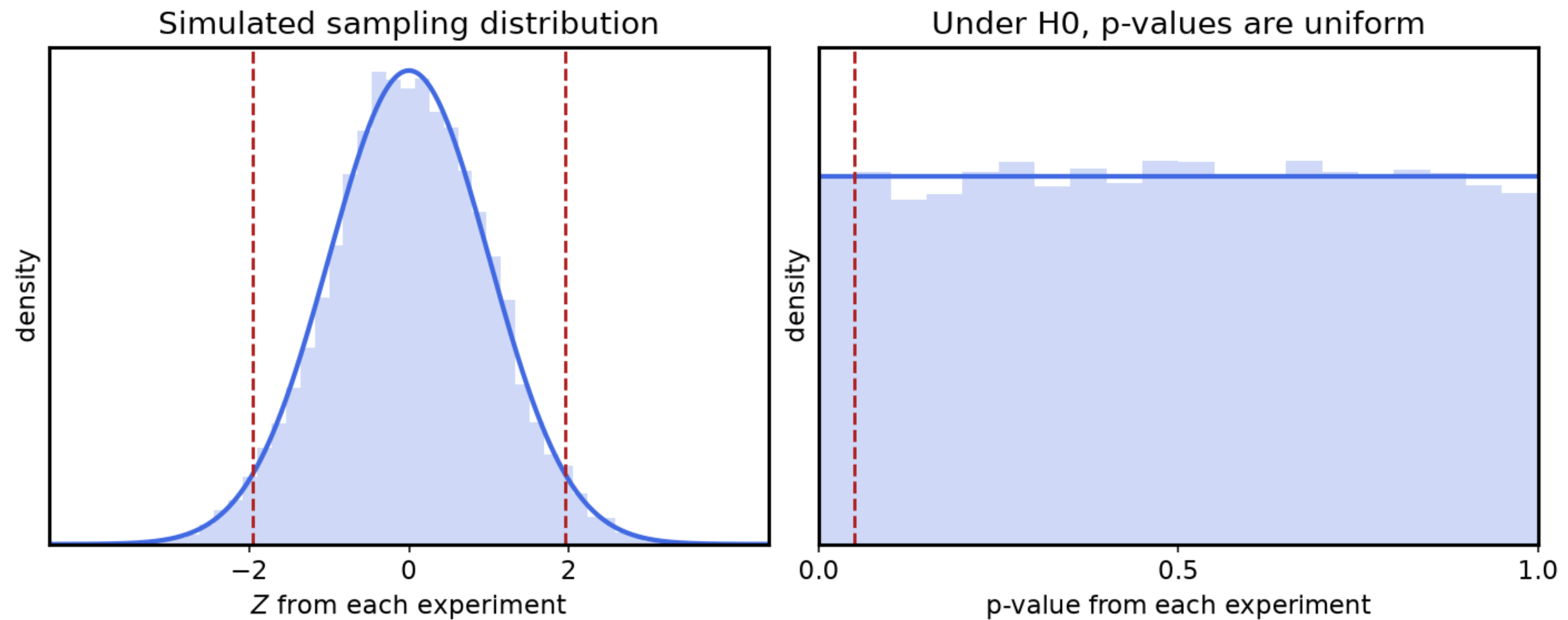
From a p -value to a decision

- Experiment: run the repeated experiments



From a p -value to a decision

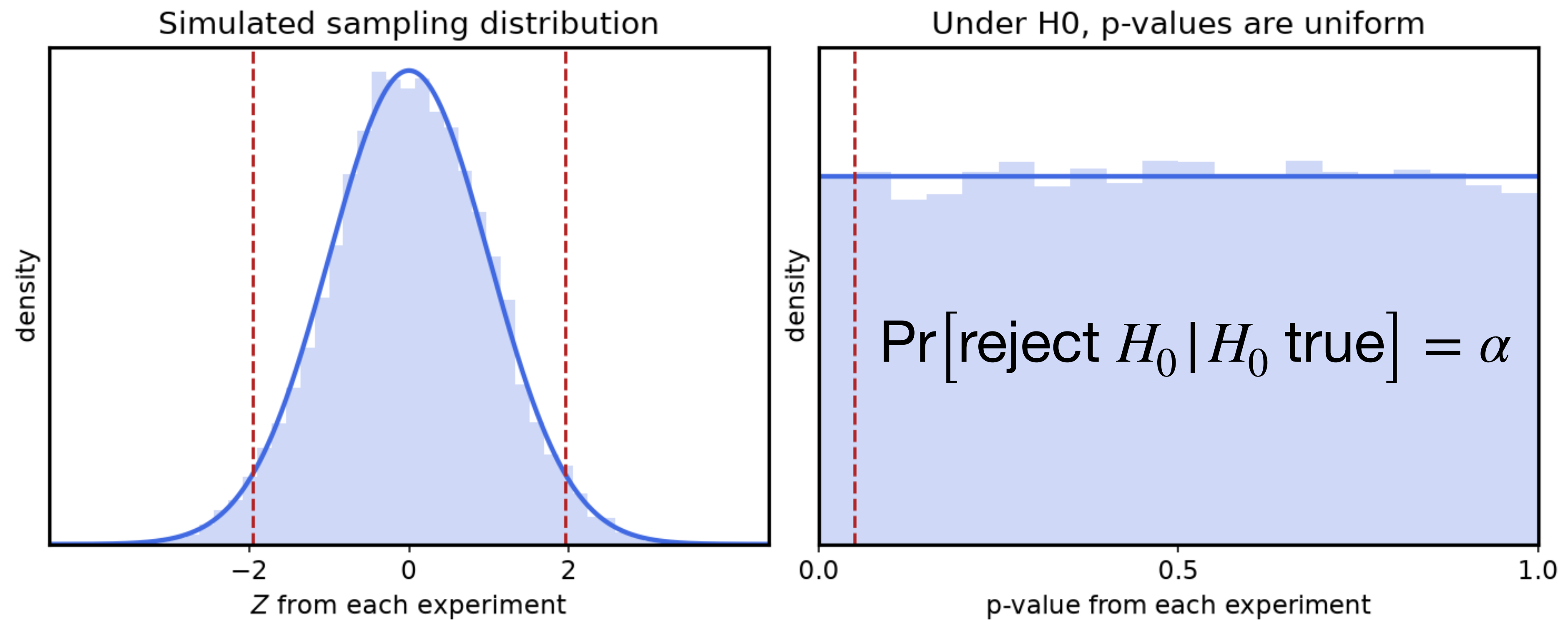
- Experiment: run the repeated experiments



- Results: we reject H_0 in about 5% of experiments.

From a p -value to a decision

- Experiment: run the repeated experiments

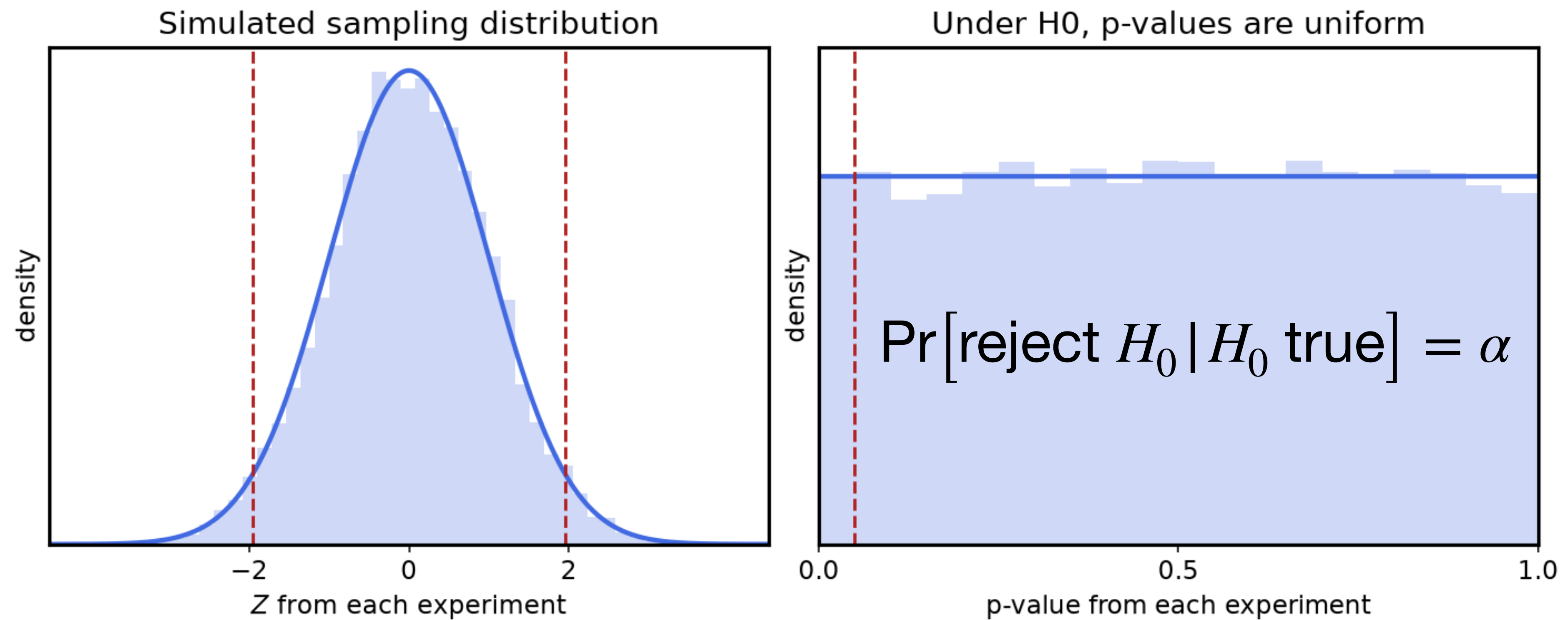


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Note: some were false rejections, caused only by noise!

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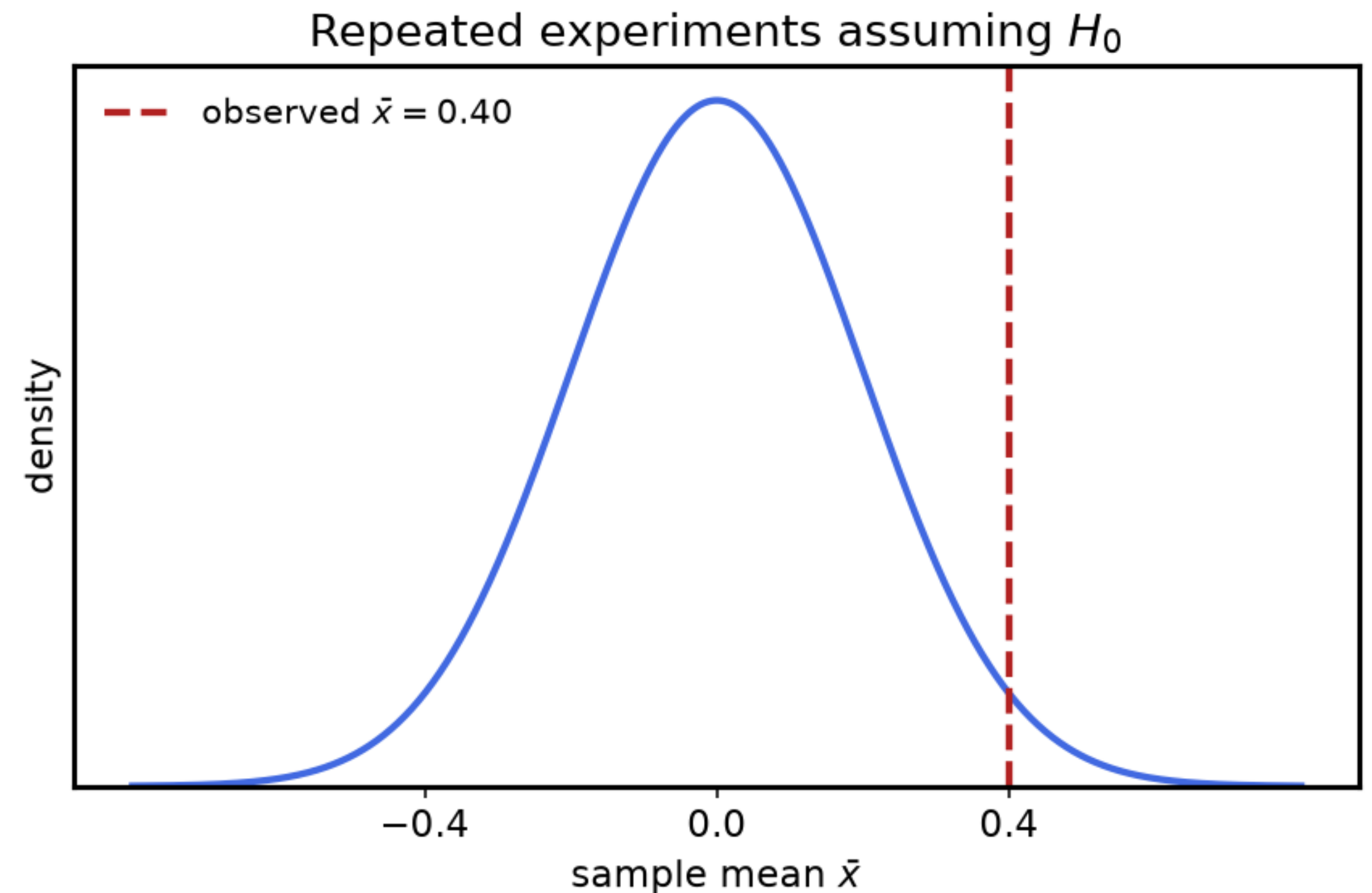
Type 1 Errors: rejecting H_0 when it is true.

Type 2 Errors: not rejecting H_0 when it is not the true theory.

-

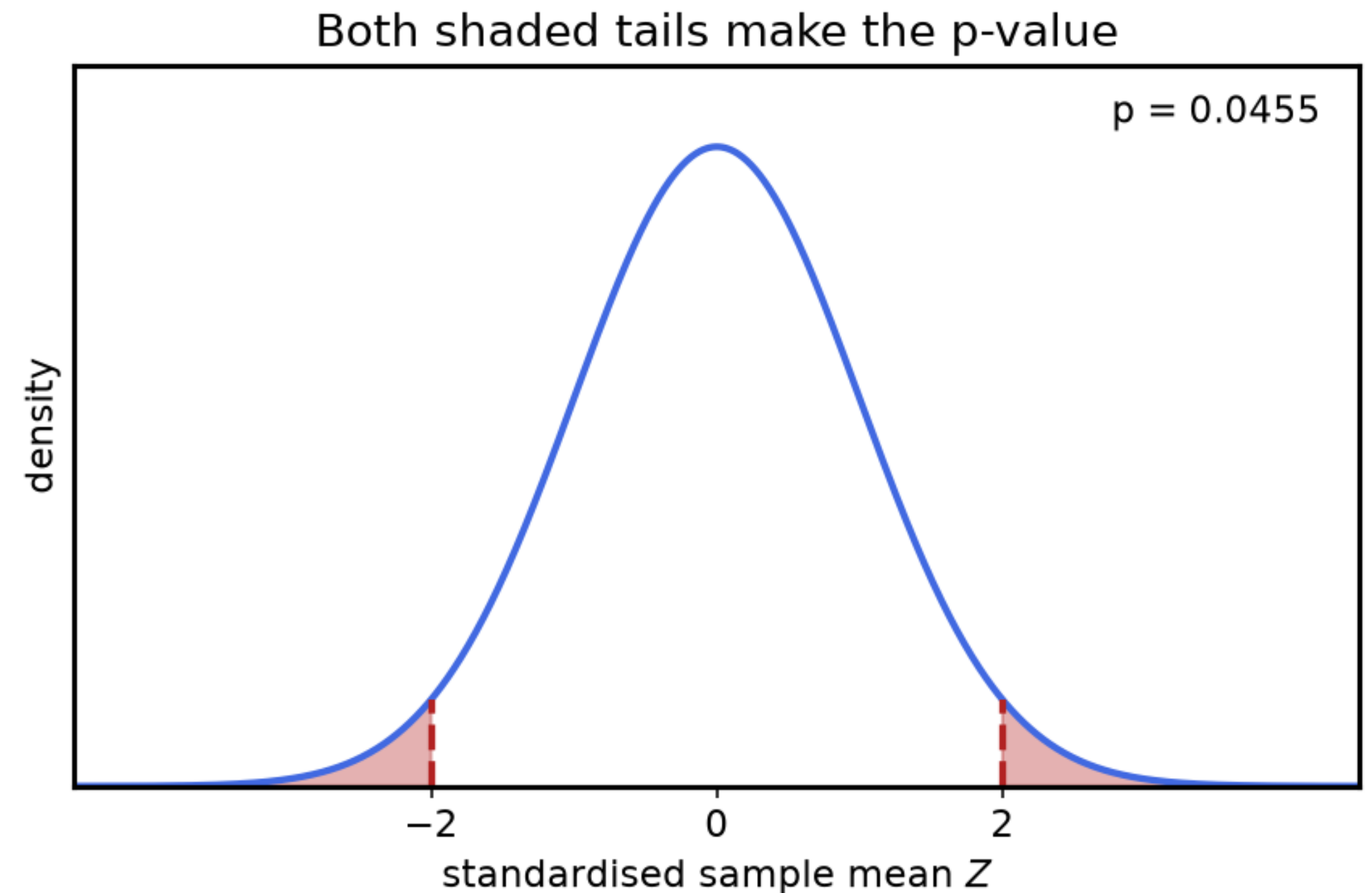
Example: the analysis you just saw

- Same experiment for $H_0 : \mu = \mu_0$
- Measure $\bar{x}_{\text{obs}}$



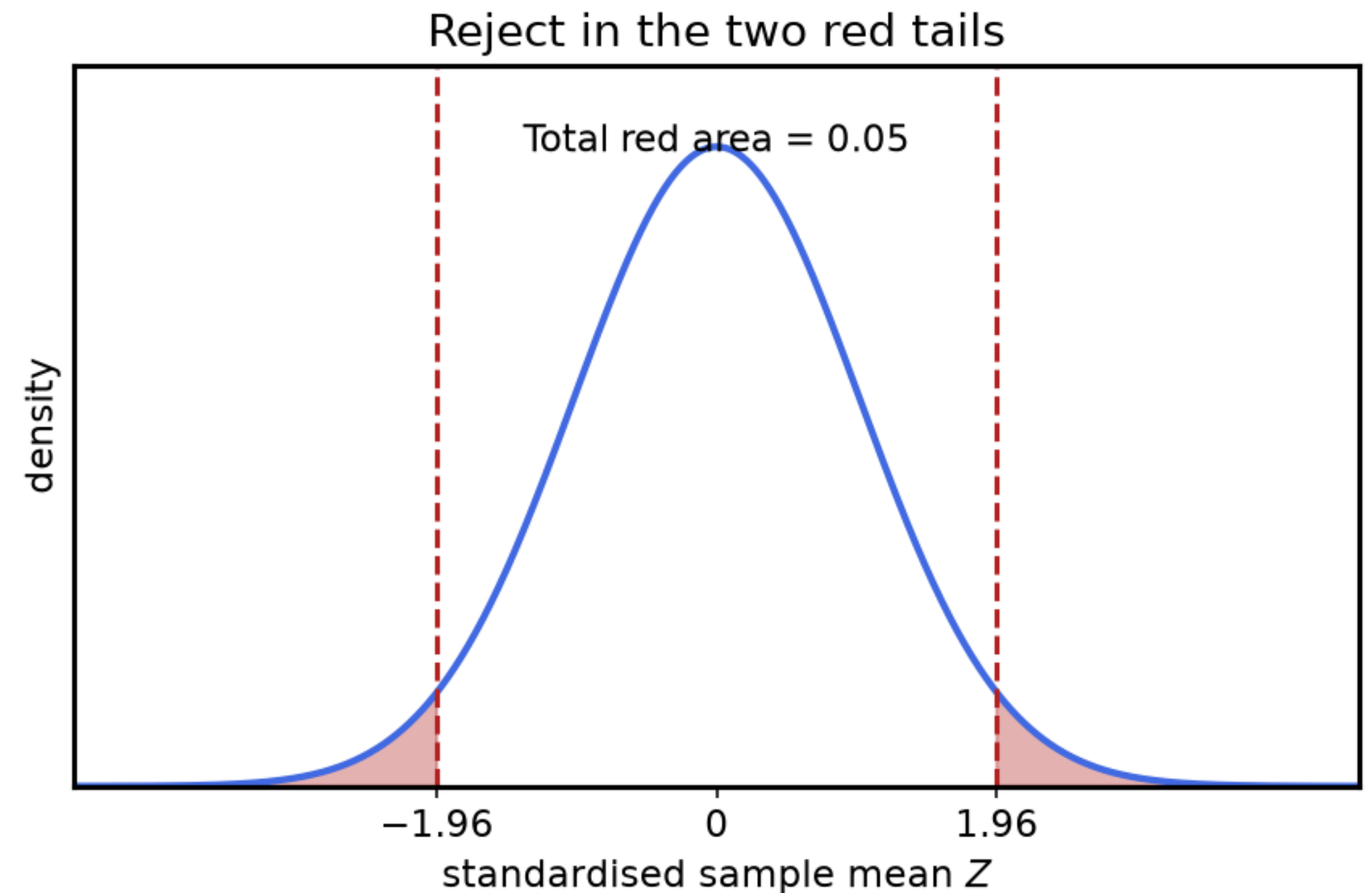
Example: the analysis you just saw

- Same experiment for $H_0 : \mu = \mu_0$
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- Calculate the p -value given $\bar{x}_{\text{obs}}$



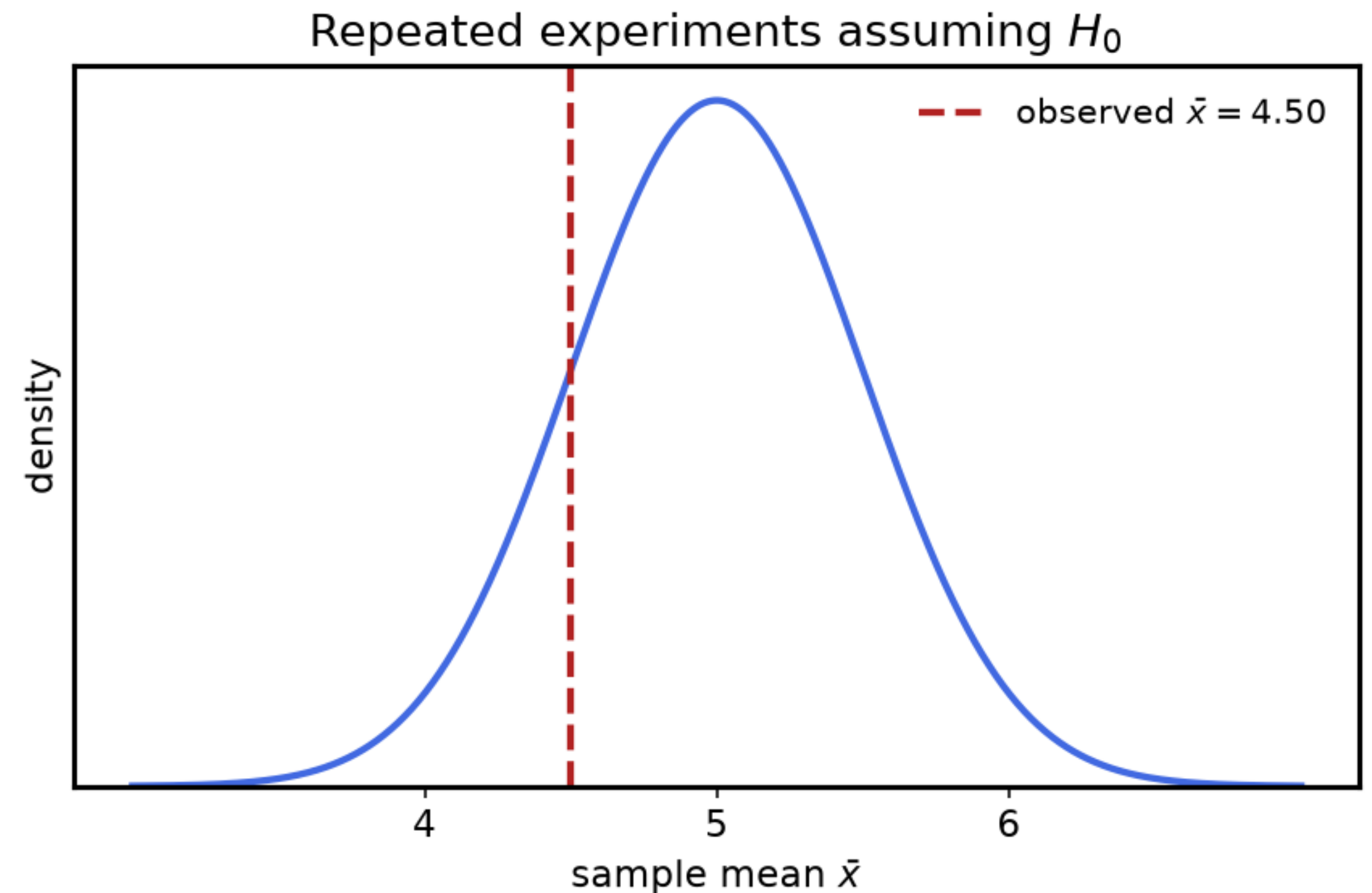
Example: the analysis you just saw

- Same experiment for $H_0 : \mu = \mu_0$
- Measure $\bar{x}_{\text{obs}}$
- Calculate the p -value given $\bar{x}_{\text{obs}}$
- Compare to same $\alpha = 0.05$ threshold
- Reject H_0 because $p \approx 0.04$



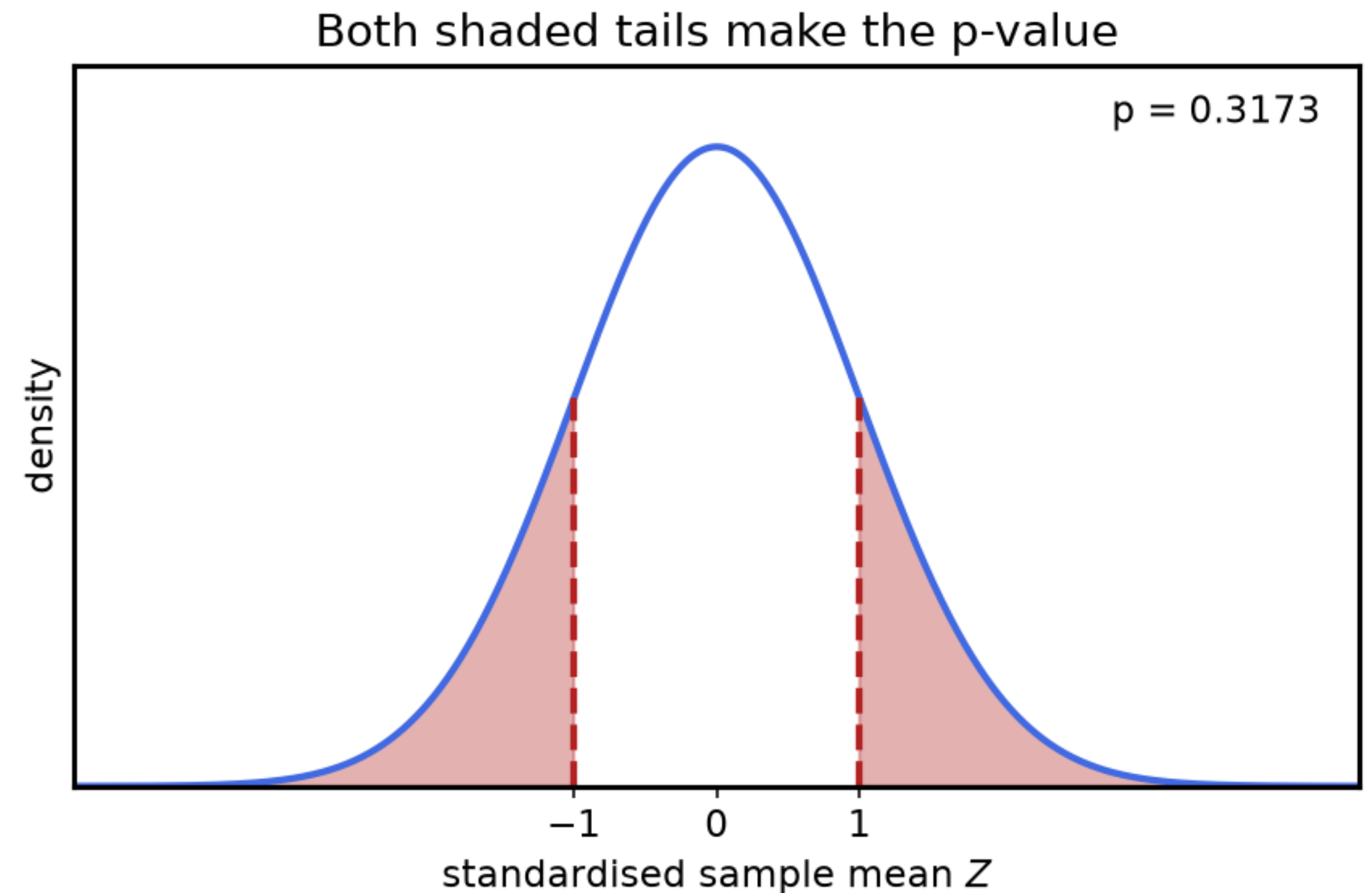
Example: Repeated analysis

- Same experiment but with different μ_0
- Measure $\bar{x}_{\text{obs}}$ again



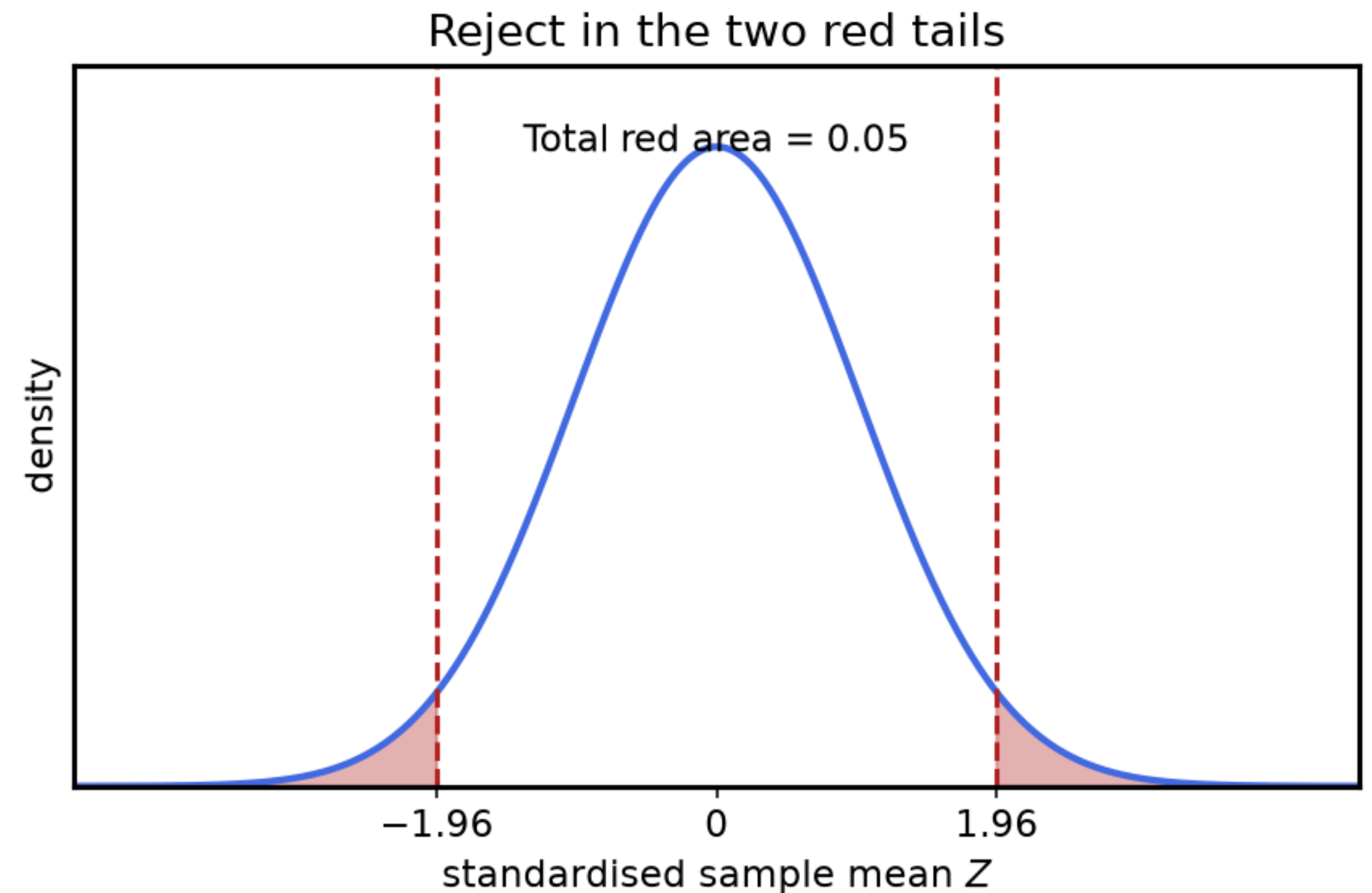
Example: Repeated analysis

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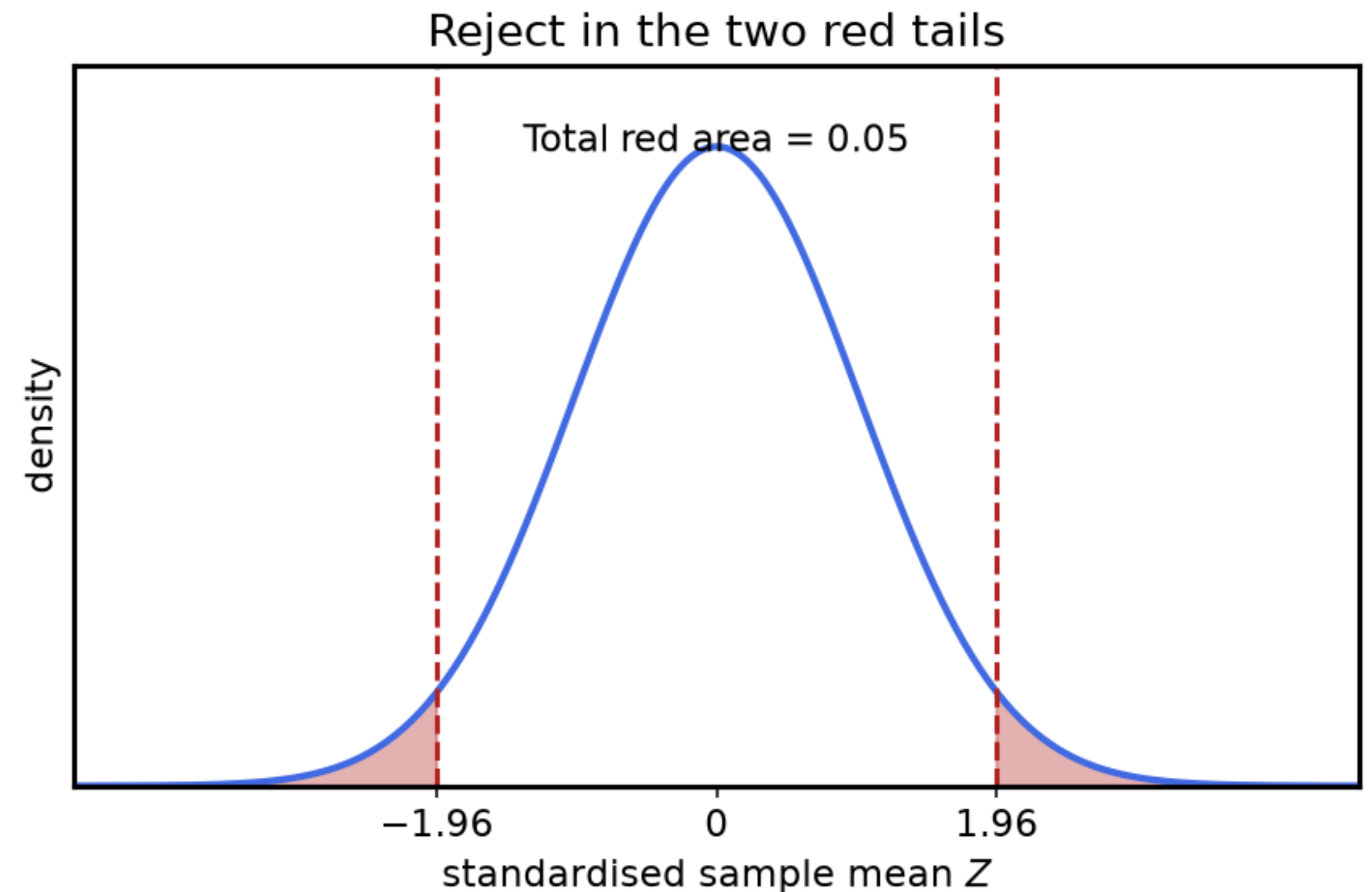
Example: Repeated analysis

- Same experiment but with different μ_0
- Measure $\bar{x}_{\text{obs}}$ again
- Calculate the p -value given $\bar{x}_{\text{obs}}$
- Compare to same $\alpha = 0.05$ threshold
- Don't reject H_0 because $p \approx 0.32$



Summary

- The frequentist uses p -values as a threshold for rejecting a model.
 - This is based on how likely it is to obtain the observed data given a model being true.
- This analysis holds for Gaussian data (trivial extension to multivariate data)
- We discussed Type-I and Type-II errors in which arise due to noise in the data and a hypothesis being false.



Write down something you learned today and
something you need to hear more about next lecture

 jed.homer@physik.lmu.de

